How many firing rates should a neuron use? Infomax staircases for a given stimulus distribution and a test in the auditory nerve

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Abstract

A sensory neuron turns a stimulus into a firing rate, and a downstream cell sees only the spike count in a short window. Barlow asked which tuning curve, the map from stimulus to rate, carries the most information about a stimulus whose distribution is given. We answer this for a neuron whose count is Poisson, with a mean between its spontaneous count and that count plus a peak count. Its most informative rate distribution uses only a few rates. We prove that the most informative tuning curve is a staircase that uses exactly these rates, with its steps at quantiles of the stimulus distribution. We prove that the number of steps grows at least as fast as 0.056 times the two-thirds power of the peak count, faster than the square-root growth of the rates that noise can tell apart. A conjectured law for the exact number, from Gaussian noise, predicted the peak counts at which the staircase gains its 21st to 25th and 47th step before they were computed, and missed its 67th. For large peak counts the square root of the optimal rate approaches a linear function of the stimulus distribution function, the Poisson form of Laughlin's histogram equalisation. We tested both predictions on gerbil auditory-nerve fibres, registering every test before analysis. Pooled over 950 fibres, rate-level functions favour the asymmetric shape predicted for natural sound levels over a symmetric sigmoid, in young and aged animals. Their counts have about a third of the Poisson variance. Under Poisson counts about half of 265 fibres use more distinct rates than the staircase allows. With counts matched to each fibre's regularity, 77 to 82 percent stay within it, against a registered 80 percent. A likelihood test finds about half within the optimum and nearly nine in ten within one step more.

Author summary

A nerve cell reports a sound to the brain with spikes, and the cells that receive its signal count only the spikes that arrive in a short time. We asked how such a cell should turn the loudness of a sound into a firing rate, given how loud the sounds of the world usually are. Barlow raised this question in 1961. We prove that the best response is a staircase instead of a smooth curve. The cell should use only a few firing rates, switch between them at loudness levels set by how common each level is, and use more rates when it can fire more spikes. We tested these ideas on gerbil auditory-nerve fibres, registering every test before looking at data. Pooled over fibres, the predicted shape fits better than a symmetric curve, in young and old animals. Treated as purely random counts, half exceed the staircase. Their counts are far more regular; with this regularity three in four to four in five stay within it, and a more sensitive test finds about half within it and nearly nine in ten within one rate more.

Index terms: efficient coding, Poisson neuron, tuning curve, channel capacity, discrete input, auditory nerve

1 Introduction

A sensory neuron turns a stimulus into a firing rate, and the cell it drives sees only how many spikes arrive in a short window. We call the map from stimulus to expected spike count the tuning curve; for sound level it is the rate-level function. If the neuron is to carry as much information as it can about the stimulus, which tuning curve should it have? Barlow asked this question in 1961 with the stimulus distribution held fixed: the neuron cannot choose what the world presents, only how it responds [1]. The tuning curve that maximizes the mutual information between stimulus and count is called the infomax tuning curve. Laughlin gave the classic answer for a neuron whose response is limited by its range but not by noise: the response should follow the cumulative distribution of the stimulus, so that every response level is used equally often [2]. Laughlin's curve is smooth; for a neuron whose count in a short window is noisy, we find that the answer is a staircase.

Spikes are noisy, and a short window holds few of them. The standard model of this noise is the Poisson count, whose variance equals its mean; we build the theory on it and then test, on recorded fibres, how much their more regular counts change its predictions. We consider a neuron whose count in the window is Poisson, with a mean that lies between its spontaneous count λ, the count with no stimulus, and $\lambda + A$. The peak count A is the largest expected count above the spontaneous one, the peak firing rate times the window. Seen as a communication channel from rate to count, such a neuron has a best distribution of rates, the one under which the count carries the most information. Rates closer together than the width of the noise are hard to tell apart from the count, so the range is best used by a few well separated rates. Information theory calls this best distribution the capacity-achieving input, and for this channel it indeed puts all its probability on a few rates [3]. Numerical work found that the infomax tuning curve of a Poisson neuron is a staircase whose number of steps grows as the window lengthens [4]; in the opposite limit of many spikes, the optimum approaches a smooth curve set by the stimulus distribution and the noise [5]. Two questions stayed open. How many steps does the optimal staircase have, as a function of the peak count? And do real neurons follow it?

This paper answers the first question with a theorem and puts the second to registered tests of two predictions: the shape of the tuning curve's envelope, and the number of distinct rates the curve uses. We show that, for any continuous stimulus distribution, the infomax tuning curve assigns the rates of the capacity-achieving input to the stimulus by its quantiles (Proposition 1). Quantiles are the stimulus values that cut its distribution into pieces of given probability, as the median cuts it into halves. The stimulus distribution therefore decides where the steps sit and nothing else: their number and their values depend only on A and λ. We prove that the number of steps grows at least like $0.056\,A^{2/3}$ (Theorem 3). A naive count of resolvable rates, the range of rates divided by the width of the noise, grows only like $\sqrt{A}$. Because the variance of a Poisson count equals its mean, the square root makes the noise nearly the same at every rate. In the coordinate $2\sqrt{r}$ the Poisson noise has unit width, and the range from 0 to A becomes an interval of length $2\sqrt{A}$, the Fisher length of the range, named after the Fisher information, which measures how precisely a count determines its rate. The optimal staircase uses more steps than that. The theorem concerns large peak counts; the fibres tested below have peak counts of about 3 to 14 (5 to 14 for those of the shape test), where the optimal staircase for Poisson counts has two to five steps. For large peak counts the square root of the optimal rate follows the distribution function of the stimulus, which is Laughlin's rule with the Poisson noise taken into account. The same problem with additive Gaussian noise of fixed variance, the Gaussian channel, has a law for its number of levels, about $0.4\,a^{4/3}$ for an input range of half-width a in units of the noise, proved from below and measured [6]. Carried over through the Fisher length, it gives a sharper law for the exact number of steps: a new step appears each time the peak count passes a threshold, and the law predicts these thresholds. The thresholds tested lie at peak counts of about 300 to 2000, far above those of neurons, so they test the mathematics of the law and not its use for neurons. We state it as a conjecture, together with the predictions it made before they were computed, including one that failed.

For the second question we used single auditory-nerve fibres of the gerbil, recorded at many sound levels with repeated presentations [7]. The auditory nerve is a natural place to look. Its rate-level functions shift towards the most frequent levels when the level distribution changes, in the midbrain and, more weakly, in the nerve itself [8]. The decibel amplitude of natural sounds has a roughly symmetric bell-shaped distribution [9], for which the theory predicts a specific asymmetric rate-level function. Every test was recorded, with its decision rule, in a dated log before the data were analysed. The log is in the repository

named under Data availability, and the entries for these tests carry a certificate of their date from an independent authority, so the order of prediction and analysis can be checked (Section 4). The failures are reported with the successes. Pooled over fibres, the rate-level functions favour the predicted asymmetric shape over a symmetric sigmoid, in young and in aged animals. Under Poisson counts about half of the fibres use more distinct rates, that is significantly different mean counts across sound levels, than the optimal staircase allows, most by one. Their counts, however, are far less variable than Poisson counts. For counts matched to the mean regularity of each fibre, more than four in five fibres stay within the optimal staircase. For counts whose regularity is matched at every rate, the model that fits them best, about three in four do, short of the registered 80%. A more powerful likelihood test finds about half of the fibres consistent with the optimal number of steps and nearly nine in ten with one step more.

Kostal and Shinomoto [10] studied a Poisson neuron with the same kind of bounds on its rate. Their downstream reader sees the full spike train, and the best input then switches between two rates. Here the reader sees only the count in a window, and counting is what makes the number of optimal rates grow with the peak count instead of staying at two.

Section 2 holds the results. Section 2.1 derives the infomax tuning curve for a given stimulus distribution. Section 2.2 states the lower bound on the number of steps, whose proof is in Appendix A. Section 2.3 states the conjectured law for the exact number and the predictions that tested it, and Section 2.4 reports the registered tests on auditory-nerve fibres. Section 3 discusses them, and Section 4 describes the data, the fits and the tests. The Supplementary Notes (S1 Text) treat counts more or less regular than Poisson counts, a spontaneous rate, codes for windows that hold few spikes, and a test for discrete levels on six further public datasets.

2 Results

2.1 Infomax tuning curves for a given stimulus distribution

Let a stimulus S have distribution P_S, and let a neuron map it to an expected spike count $r(S)$ in one counting window, with $\lambda \le r(s) \le \lambda + A$; here λ is the spontaneous count and A the peak count above it. The response is $Y \mid S = s \sim \text{Poisson}(r(s))$. The tuning curve r is the design variable and P_S is fixed, as in Barlow's and Laughlin's formulation of efficient coding. Let $C(A, \lambda)$ be the capacity of the Poisson channel with input in $[\lambda, \lambda + A]$, the largest mutual information between rate and count that any distribution of rates in this range can give; for example, $C(3, 0) = 0.5944$ nats, or 0.858 bits per window. Let $\pi = \sum_{j=1}^{K} w_j \delta_{x_j}$, with weights w_j and mass points $x_1 < \cdots < x_K$, be its capacity-achieving input: the distribution of rates that gives probability w_j to the rate x_j and reaches the capacity. Shamai proved that it is unique and has finitely many mass points [3]. The letter π always denotes this distribution; the number 3.14159 enters only inside constants of Theorem 3 and its proof. Write $W_j = w_1 + \cdots + w_j$ and $W_0 = 0$.

Proposition 1. *For every measurable tuning curve r with values in $[\lambda, \lambda + A]$,*

$$I(S; Y) \le C(A, \lambda),$$

with equality if and only if the law of $r(S)$ is π. If P_S has a continuous distribution function F_S, equality is attained, and, up to a P_S-null set, the nondecreasing maximizer is the staircase

$$r^*(s) = x_j \qquad \text{for } F_S(s) \in (W_{j-1}, W_j], \quad j = 1, \ldots, K.$$

Every maximizer takes exactly K values outside a P_S-null set.

In words: no tuning curve carries more information than the capacity, and a tuning curve reaches it exactly when the rates it produces have the distribution π. The law of $r(S)$ is this distribution of produced rates, and a P_S-null set is a set of stimuli of probability zero, on which two tuning curves may differ without any change in information. The count depends on the stimulus only through the rate, so passing from the stimulus to the rate loses nothing, and the information of the tuning curve is the information of its rates. To build the nondecreasing one, sort the stimuli by their cumulative probability. Give the lowest rate x_1 to the

lowest fraction w_1 of stimuli, the next rate x_2 to the next fraction w_2, and so on up to x_K. Any assignment of the rates to pieces of the stimulus distribution with these probabilities is optimal; the nondecreasing one has the shape of the rate-level functions recorded in the auditory nerve.

Proof. Put $X = r(S)$. The count is drawn from the rate alone, so given X the count does not depend on S; thus $S \to X \to Y$ is a Markov chain and $I(S;Y) \le I(X;Y)$. Since X is computed from S, also $X \to S \to Y$ is a Markov chain and $I(X;Y) \le I(S;Y)$. Hence $I(S;Y) = I(X;Y)$. The variable X takes values in $[\lambda, \lambda + A]$, so $I(X;Y) \le C(A,\lambda)$, with equality if and only if the law of X is a capacity-achieving input, that is π by uniqueness. If F_S is continuous, $U = F_S(S)$ is uniform on $(0,1)$. Let G be the distribution function of π and $G^{-1}(u) = \inf\{x : G(x) \ge u\}$ its quantile function. Then $r^*(S) = G^{-1}(U)$ has law π, so r^* attains the bound; it is nondecreasing and takes the value x_j exactly when $U \in (W_{j-1}, W_j]$. Any maximizer has law π, a K-point law, so it takes K values almost surely. □

The proposition separates two questions. The stimulus distribution decides where the steps sit, at its quantiles W_j; it does not change their number or their values, which depend only on A and λ. A staircase shape of the optimal tuning curve was found numerically for Poisson neurons with a finite coding window [4]; the proposition derives it for every continuous stimulus law from the uniqueness and discreteness of π. Fig 1 shows the construction for a Gaussian stimulus.

Fig 1. The infomax tuning curve of a Poisson neuron is a staircase. (A) A Gaussian stimulus distribution; the three dashed lines, W_1 to W_3 from left to right, mark the stimulus values below which the stimulus falls with probability W_j; the optimal tuning curve steps there for peak count $A = 10$ and no spontaneous count. (B) The optimal tuning curves for $A = 3$, 10 and 25, as a fraction of the peak count, with 2, 4 and 5 steps. Much of the weight of the optimal input sits at the lowest rate, silence, and at the peak rate, so the inner steps lie close to the middle of the stimulus range; the dotted curve $F_S(s)^2$ is the envelope that the staircase approaches only for large A. (C) The number of steps against the peak count, from numerical solutions of the capacity problem, with the asymptotic lower bound $0.056\,A^{2/3}$ of Theorem 3; at these peak counts the bound lies far below the number of steps, since it concerns how the number grows for large A. All values are computed numerically.

Corollary 2. *Fix $\lambda \ge 0$, and let $K(A,\lambda)$ be the number of mass points of π at peak count A. Then the infomax tuning curve of Proposition 1 has $K(A,\lambda)$ steps, and* $\liminf_{A\to\infty} K(A,\lambda)/A^{2/3} \ge 0.0560$.

Proof. The number of values of r^* is the number of mass points of π; apply Theorem 3, stated in the next section. □

The corollary holds for every fixed spontaneous count, as the fibres have. At the small peak counts of the fibres the bound says little, and the number of steps is computed numerically, as in Section 2.4.

In the coordinate $2\sqrt{r}$ the Poisson noise has unit width and the range $[0, A]$ has length $2\sqrt{A}$, its Fisher length. A count of distinguishable levels, the Fisher length divided by the width of the noise, therefore grows only like $\sqrt{A}$. Since $2/3 - 1/2 = 1/6$, the infomax staircase uses more steps than that, by a factor of order $A^{1/6}$ at least. This comparison concerns large A: the constant 0.056 is small, and at the peak counts of Fig 1C the bound lies far below the actual number of steps.

The large-A limit. As A grows, the steps become dense. In the limit of small noise the capacity-achieving distribution of the rate approaches the density proportional to the square root of the Fisher information. The Fisher information of a Poisson count about its rate r is $1/r$, so this density is proportional to $r^{-1/2}$. McDonnell and Stocks [5] use this argument to derive the optimal tuning curve as a nonlinear function of the stimulus distribution function. Under this density $\sqrt{r}$ is uniform on $[\sqrt{\lambda}, \sqrt{\lambda + A}]$. Proposition 1 maps the quantiles of the stimulus to the quantiles of this law, so that

$$r(s) \approx \left(\sqrt{\lambda} + (\sqrt{\lambda + A} - \sqrt{\lambda})\, F_S(s)\right)^2. \tag{1}$$

Without spontaneous count Eq (1) becomes $A\,F_S(s)^2$, and it is close to $\lambda + A\,F_S(s)^2$ only when λ is small compared with A. This is histogram equalization in the square-root coordinate, the Poisson form of

Laughlin's rule. The approximation describes the envelope of the optimal staircase when A is large and the steps are many. At the peak counts of the rate-level functions of Section 2.4, about 5 to 14 for four fibres in five, the staircase has only a few steps, and the smooth curve is the predicted shape of its envelope. The shape test of Section 2.4 checks this envelope, and its discreteness test checks the number of steps.

2.2 The number of firing rates grows at least like $A^{2/3}$

Let $K(A,\lambda)$ be the number of mass points of the capacity-achieving input of the channel $Y \sim \mathrm{Poisson}(X)$ with $\lambda \le X \le \lambda + A$, where $\lambda \ge 0$ is the spontaneous count, and $K(A) = K(A,0)$. The constant of the theorem comes from a one-dimensional optimization over a parameter ξ of the proof; a reader interested only in the result may skip the definitions that follow. In words, c_P below is the best constant that the construction of the proof yields, and λ_M and λ_ℓ are numbers of that construction, unrelated to the spontaneous count λ. For $\xi \in (0,1)$ put $\phi_\xi = \xi \log \xi + 1 - \xi$, let $\kappa_L(\xi) \in (0,\xi)$ be the root of $\log k + (1-k)/\xi = 0$, and put

$$h(f) = \frac{\xi \log f + 1 - f}{\phi_\xi}, \qquad \mathcal{L}(1,z) = \frac{z-1}{\log z}, \qquad \varrho = \frac{\mathcal{L}(1,M)}{\mathcal{L}(1,\ell)},$$

where $M = e^{\lambda_M}$ with $\lambda_M = 1 - \psi(2) + e^{\psi(2)}$, $\ell = e^{-\lambda_\ell}$ with $\lambda_\ell = 2.777$, and ψ is the digamma function. Let

$$c_0^{\max}(\xi) = \Big[\big(1 + \xi\kappa_L^{-1/2}\big)\kappa_L^{-1/4}\varrho^{1/2}/\phi_\xi\Big]^{-2/3}, \qquad \mathcal{I}(\xi) = \int_{\kappa_L}^{1} \Big(\frac{h(f)}{f}\Big)^{1/2} df.$$

Both quantities enter the bound as follows.

Theorem 3. *For every fixed $\lambda \ge 0$,*

$$\liminf_{A\to\infty} \frac{K(A,\lambda)}{A^{2/3}} \;\ge\; c_P := \max_{0<\xi<1} \frac{c_0^{\max}(\xi)^{1/2}\,\mathcal{I}(\xi)}{2\pi} \;>\; 0.0560.$$

At $\xi = 0.2411$, near the maximizer, $c_0^{\max} = 0.06491$, $\mathcal{I} = 1.38249$ and the bound is 0.05606.

In words: for every $\varepsilon > 0$ and every large enough A, the optimal input has at least $(c_P - \varepsilon)\,A^{2/3}$ mass points; the bound concerns growth, and the lower limit lim inf allows finitely many exceptions at small A. The proof is given in Appendix A; its constants are enclosed in interval arithmetic, so the numerical steps of the proof are proved bounds. In the Fisher half-length $a = \sqrt{A}$ the bound reads $K \ge (c_P - o(1))\,a^{4/3}$, where $o(1)$ tends to zero as A grows. This is the exponent of the count law of the Gaussian channel: the companion paper [6] proves that the Gaussian channel with input in $[-a,a]$ has at least $0.166\,a^{4/3}$ mass points for large a, and measures about $0.4002\,a^{4/3}$. The best published lower bound on the same number $K(A)$ that we know of is of order $\sqrt{A}$ [11, 12], so the theorem raises the proved exponent from 1/2 to 2/3. The best published upper bound is of order A [12], so the proved order of growth now lies between $A^{2/3}$ and A; the law of Section 2.3 places it at $A^{2/3}$. The constant of Theorem 3 is far from sharp. At the peak count 2000 the bound gives about 9 steps, while the optimal input has 66; the theorem gives the order of growth from below and leaves the exact count to the law of Section 2.3.

2.3 A conjectured law for the number of steps

Theorem 3 fixes the order of growth from below. The exact number of steps is described by a law that is conjectured and tested by predictions recorded before the computations. In the coordinate $t = 2\sqrt{x}$ the Poisson noise has nearly unit variance, so the range $[0,A]$ has Fisher length $2\sqrt{A}$ and half-length $\sqrt{A}$. The levels of the optimal input are spaced like those of the Gaussian channel with unit noise and input in $[-a,a]$, here with $a = \sqrt{A}$ [6]. The exception is the level at silence: there the output is degenerate, and the levels next to it form a boundary layer. Let $A_g(K)$ be the half-length at which the Gaussian count changes from $K-1$ to K. Let A_K be the peak count at which the Poisson count does. The law reads

$$\sqrt{A_K} = A_g(K) + c(K), \qquad c(K) = \tfrac{1}{2}\,E\big(g_{\mathrm{eff}}(K)\big) + 0.002, \qquad g_{\mathrm{eff}}(K) = A_g(K+1) - A_g(K-1). \qquad (2)$$

Table 1. Count changes of the Poisson channel predicted by the law (2) before they were computed. A_K is the peak count at which the optimal input gains its K-th mass point, without spontaneous count. The tolerance of 0.15 in A was fixed with the first prediction, before any of these changes was computed; since a change δ in A moves $\sqrt{A}$ by about $\delta/(2\sqrt{A})$, it is $0.15/(2\sqrt{A})$ in the half-length, 0.0037 at $K = 25$ and 0.0017 at $K = 67$, close to the scatter of the law; relative to A it is 3.6×10^{-4} at $K = 25$ and 7.5×10^{-5} at $K = 67$. Row 21 was predicted from an earlier local fit of the same law and row 22 from the law without the correction 0.002; the later rows use the finite-chain variant as written.

K	predicted A_K	computed A_K	difference	outcome
21	307.99	308.0874	+0.0974	held
22	333.14	333.1959	+0.0559	held
23	358.97	358.9655	−0.0045	held
24	385.39	385.3810	−0.0090	held
25	412.43	412.4271	−0.0029	held
47	1145.508	1145.6157	+0.1077	held
67	2001.381	2001.5878	+0.2068	failed

In words, the K-th step of the neuron appears where the Gaussian channel gains its K-th level, measured in the Fisher half-length, shifted by a small wall constant $c(K)$ that accounts for the boundary layer at silence. To predict A_K one takes $A_g(K-1)$, $A_g(K)$ and $A_g(K+1)$ from the Gaussian channel, forms $g_{\text{eff}}(K)$, evaluates $c(K)$ and squares $A_g(K) + c(K)$. Here $E(g)$ is an excess defined from two half-line chains, small models of the levels next to one end of the range. Each chain has thirty free levels next to the wall and frozen levels at spacing g beyond them, one chain with Poisson noise and one with Gaussian noise. The excess is the difference between their last free levels, measured in the Fisher coordinate. We call this the finite-chain variant. The fit $E(g) = 0.9840 - 0.2347\,g$ to the chain's own excess on $1.43 \le g \le 2.35$ never sees a Poisson count change, so the rows of Table 1 are predictions out of sample. The term 0.002 is a correction for the finite length of the chain, estimated from chains of 20, 30 and 40 levels. A second variant, the infinite-chain variant, uses the fit for the limit of an infinitely long chain, $E(g) = 1.0049 - 0.2454\,g$ without the correction. As a check after the fact, over $K = 5$ to 24 the term $\frac{1}{2}E(g_{\text{eff}})$ alone matches the measured $c(K)$, which runs from 0.237 to 0.335, with a root-mean-square deviation of 0.0015, about half a percent. How the count changes are computed is described in Section 4.

Table 1 lists the seven predictions made so far. Six held. The prediction for the 67th level failed: the change lies 0.207 above the predicted value, outside the tolerance. Its miss and that at $K = 47$ grow with K, and both lie where g_{eff} is 1.04 to 1.14, below the range on which the chain fit was made. The miss at $K = 67$ is 1.0×10^{-4} of A. The infinite-chain variant lies 0.05 above the measured changes at $K = 47$ and $K = 67$, a nearly constant offset, while the misses of the finite-chain variant grow as g_{eff} falls. Read after the failure, this points to the finite-chain correction, estimated from chains of 20 to 40 levels, carried outside its range; it does not single out the form of the law. The law is a conjecture that holds to 0.15 in A up to 47 levels and fails at 67. The computed count changes themselves are being turned into proved statements. At the lower end of the bracket of the 25th level, $A = 412.42711264916346$, a computer-assisted proof shows that the optimal input has exactly 24 mass points (Section 4.6). Three further changes were predicted with both variants before any computation, the finite-chain value first and the infinite-chain value second: A_{26} at 440.085 and 440.141, A_{48} at 1184.393 and 1184.557, and A_{68} at 2048.241 and 2048.503. They will decide between the variants and are not yet computed. These peak counts lie far above those of real fibres, whose staircases have only a few steps. The predictions test the mathematics of the law, while the fibres of Section 2.4 test its consequences for neurons.

Through the Fisher length the law carries the growth of the Gaussian count to the neuron. The Gaussian count grows like $a^{4/3}$ at half-length a (proved from below and measured in [6]), and with $a = \sqrt{A}$ this becomes a Poisson count of order $A^{2/3}$, the order that Theorem 3 proves from below.

2.4 Auditory-nerve fibres

The data are the public recordings of single auditory-nerve fibres of the gerbil by Heeringa and colleagues [7]. Two of their protocols serve our questions, and each test uses the recordings of one protocol. In rate-level

recordings, tones at the best frequency, the frequency the fibre responds to most, are played at up to 27 sound levels with 10 presentations each; these serve the test of shape. Phase-locking recordings were made in fibres with best frequencies below about 5 kHz, where spikes lock to the phase of the tone. In them, tones are played at 5 to 8 sound levels with 20 to 40 presentations each; the larger number of presentations serves the test of discreteness, which compares the mean counts of neighbouring sound levels. Counts are taken in a window of 50 ms from tone onset. Every test below was fixed, with its decision rule and its controls on synthetic fibres, before the data were analysed; Section 4 gives the details. Table 2 lists the tests and whether their registered rule held.

Shape. The decibel amplitude of natural sounds has a roughly symmetric bell-shaped distribution [9]. Take the distribution of sound level L to be Gaussian with mean μ and spread σ, so that $F_S(L) = \Phi((L-\mu)/\sigma)$, where Φ is the standard normal distribution function. For a small spontaneous count r_0, the fitted value of λ, the large-A form of Section 2.1 becomes $r_0 + A\,\Phi((L-\mu)/\sigma)^2$, an asymmetric sigmoid in L. The square of the S-shaped Φ rises from its floor more steeply than it approaches its ceiling, unlike a symmetric sigmoid (Fig 2A). This is the form the test registered. We compared this predicted shape with the symmetric sigmoid, the usual description of rate-level functions, which has the same four parameters (spontaneous count, peak count, centre and width), by Poisson likelihood on the single-trial counts of 950 fibres. A single fibre at ten presentations distinguishes the two shapes only weakly, so the decision pooled the fibres. For each fibre, d is the log-likelihood of the predicted shape minus that of the sigmoid, in nats over all its single-trial counts. The mean of d over fibres was 0.2027, with a bootstrap lower bound of 0.1495. Only 50.8% of fibres favour the predicted shape individually, so the pooled preference comes from a minority of fibres with a large difference (Fig 2B). The decision rule asked for a positive pooled mean and a median spread σ between 6 and 20 dB, not for a majority of fibres. Fibres of one animal need not be independent; a bootstrap that draws the 103 animals with all their fibres, a reading made after the test, gives a lower bound of 0.1466. In a split made after the test, the preference was strongest in fibres with large peak counts and in fibres with a low spontaneous count (below 0.5 per window), 78.9% of which favoured the predicted shape. It held separately in young-adult and in quiet-aged animals, aged animals raised in a quiet environment so that noise damage to the cochlea is kept small [7], with lower bounds of 0.1427 and 0.0768. The spread σ that the fitted shapes imply for the level distribution has a median of 8.53 dB. From recordings of natural sounds, Escabí and colleagues measured standard deviations of the decibel amplitude in frequency bands of 9.1 dB for environmental sounds, 13.3 dB for vocalizations and 14.8 dB for speech, so the fitted spread lies just below the range of natural sounds.

Section 2.1 also gives the exact form for any spontaneous count, $\left(\sqrt{r_0} + (\sqrt{r_0 + A} - \sqrt{r_0})\,\Phi\right)^2$. In the registered form the spontaneous count is added to the stimulus-driven count; in the exact form the two mix inside the square. The prediction of the exact form failed: a second registered test found that it describes the fibres worse than the registered one, with a mean log-likelihood difference of -0.0904 per fibre and a bootstrap lower bound of -0.1062, although it still beats the symmetric sigmoid. The fibres therefore behave as if their spontaneous count were an offset added to the stimulus-driven count. A third registered test compared the predicted form with the complementary log-log curve $r_0 + A(1 - e^{-e^{(L-\mu)/\sigma_s}})$, which has four parameters and the opposite direction of asymmetry. The predicted form fitted better: the mean of d was 1.2463, with a bootstrap lower bound of 1.1048, and 80.2% of fibres favoured it. Against the Gompertz curve $r_0 + A\,e^{-e^{-(L-\mu)/\sigma_s}}$, whose asymmetry has the predicted direction and is stronger, the predicted form fitted worse, with a mean difference of -0.1834 and a lower bound of -0.2432; this comparison is a reading, not a registered rule. The fibres thus show the predicted direction of asymmetry, while the predicted form is not the best curve of that direction. Fig 2A shows a fibre chosen by a rule fixed before plotting: among the 50 fibres whose fitted peak count is closest to the median, the one whose difference d is the median.

Discreteness. Proposition 1 predicts that an optimal fibre shows at most K different mean counts across any set of sound levels, where K is the number of steps of the optimal staircase at the fibre's peak and spontaneous counts. Measured mean counts are noisy, so two neighbouring sound levels count as having different mean counts only when a test finds their counts significantly different. We counted the different mean counts D in this way, with an exact binomial test between neighbouring levels and Holm's correction for multiple comparisons [13] within each fibre (Section 4). With K computed for Poisson counts, the bound $D \le K$ held in 51.7% of the 265 fibres that met the selection rule of Section 4, against the registered 80%.

Table 2. Registered tests on auditory-nerve fibres. d: log-likelihood of the predicted shape minus that of the symmetric sigmoid, in nats per fibre; LB: bootstrap 95% lower bound of the mean of d; σ: spread of the level distribution implied by the fitted shape; D: number of significantly different mean counts across the tested sound levels; K: number of steps of the optimal staircase at the fibre's peak and spontaneous counts, for the count model named in the row; drift: a slow change of rate across presentations; Fano factor: variance of the count divided by its mean. The fibres of the two age groups are 881 of the 950; the other 69 come from middle-aged animals and were not part of the registered replication.

test	fibres	registered rule	result
shape (bell-shaped level law)	950	LB > 0 and median σ in $[6, 20]$ dB	LB 0.1495, σ 8.53 dB: held
shape, young-adult animals	608	LB > 0	LB 0.1427: held
shape, quiet-aged animals	273	LB > 0	LB 0.0768: held
exact square-root shape against the registered shape	950	LB of the mean log-likelihood gain > 0	LB -0.1062: did not hold
shape against the curve of opposite asymmetry	950	LB of the mean log-likelihood gain > 0	LB 1.1048: held
staircase, Poisson counts, 50 ms	265	$D \leq K$ in at least 80% of fibres	51.7%: did not hold
drift explains the excess	265	drift in at least 30% of fibres	5.7%: did not hold
staircase, Poisson counts, 75 ms	271	$D \leq K$ in at least 80% of fibres	55.7%: did not hold
counts as variable as Poisson counts	265	median Fano factor in $[0.8, 1.2]$	median 0.355: did not hold
staircase, binomial counts, 50 ms	265	$D \leq K$ in at least 80% of fibres	88.3%: held
staircase, counts matched to each fibre's Fano factor, 50 ms	265	$D \leq K$ in at least 80% of fibres	82.3%: held
staircase, counts matched to each fibre's Fano factor at every rate, 50 ms	265	$D \leq K$ in at least 80% of fibres	77.0%: did not hold
staircase, likelihood statistic, count matched at every rate, 50 ms	265	at least 80% of fibres consistent with at most K steps	40.4%: did not hold
the same, with the range from the spontaneous count	265	the same	49.4%: did not hold

Fig 2. The rate-level functions and the predicted shape. (A) One fibre, chosen by a rule fixed before any plot was made: among the 50 fibres whose fitted peak count is closest to the median over all fibres, the one with the median d. Grey points: single-trial counts in 50 ms; black points: mean count per sound level; solid line: fitted predicted shape; dashed line: fitted symmetric sigmoid; step line: the staircase of Proposition 1, predicted and not fitted, for a Gaussian level law with the fitted mean and spread, at the fibre's fitted spontaneous count 5.46 and peak count 9.25 above it, where the capacity-achieving input has two rates. The staircase is what an optimal fibre with these counts would show; the data are compared with it through the number of distinct mean counts, not point by point, and Fig 3C shows that comparison for one fibre of the discreteness test. For this fibre the two fitted shapes nearly coincide ($d = -0.035$). (B) The difference d for all 950 fibres, with zero, the mean 0.203 and the median 0.0015 marked; the median near zero shows that the pooled advantage comes from a minority of fibres; one value below -2 and ten above 4 are drawn at the ends.

Most excesses were of one rate: the difference $D - K$ was -1 in 34 fibres, 0 in 103, $+1$ in 103 (38.9%), $+2$ in 23 and $+3$ in 2. A fully graded fibre would show as many distinct mean counts as tested sound levels; only 8.7% of fibres did, so the fibres are not fully graded either. Slow drift, a change of rate across the presentations of one sound level, does not explain the excess: only 5.7% of fibres drifted, close to the 4.5% that the drift test marks in stationary synthetic fibres. Fibres with an excess drifted no more often than the others. The recorded trials of most fibres end near 80 ms after tone onset, so 75 ms is the longest window inside every trial; it raised the fraction only to 55.7%, with 271 fibres meeting the selection rule at that window.

The counts, however, are not Poisson. The tests that follow were registered after the Poisson test had failed and in response to it. Each had its rule and its bar of 80% fixed before it was run (Section 4). The Fano factor of a count, its variance across presentations divided by its mean, is 1 for Poisson counts. Over the 1430 sound levels of the 265 fibres with a mean count of at least one, its median was 0.355, with quartiles 0.259 and 0.524, and 86.0% of the levels lay below 0.67. A count that varies less than a Poisson count has less noise at each rate, so nearby rates are easier to tell apart, the optimal input of its counting channel can hold more rates, and the K above is too small for these fibres. The proof of Proposition 1 uses only that the count depends on the stimulus through the rate: for any counting channel, the staircase built from a capacity-achieving input with finitely many mass points is an optimal tuning curve. We therefore recomputed K for a binomial count with n trials and mean r, whose Fano factor is $1 - r/n$. The binomial count is the simplest count less variable than a Poisson one: a neuron that fires at most once in each of n short bins of the window produces it. For a peak count of 10 in the 50 ms window, n is 10 and each bin is 5 ms long, much longer than the absolute refractory period of about 0.6 ms of these fibres [7]. The bins are a device of the model: the binomial count describes how regular the counts are, not the time course of recovery. Since the count takes only $n + 1$ values, a capacity-achieving input with at most $n + 1$ mass points exists [14]. For n we took the smallest integer at or above $\lambda + A$, the most regular binomial count that reaches the fibre's peak rate. With this K the bound $D \leq K$ held in 88.3% of the 265 fibres, above the registered 80%; five fibres with a Fano factor of at least one kept Poisson counts. The binomial K exceeded the Poisson one by one rate in 177 fibres and by two in 28. The binomial count models the regularity of the fibres without being fitted to it: at the fibre's mean count it is more regular than the fibre in 158 fibres and less regular in 102. A second registered test therefore matched the count to each fibre. It used a Conway-Maxwell-Poisson count [15], whose dispersion is set by one parameter ζ: the value 1 gives Poisson counts, and larger values give less variable ones. For each fibre ζ was chosen so that the Fano factor at the fibre's mean count equals the measured one. With this K the bound held in 82.3% of the fibres, again above the registered 80%. For this count, too, the capacity-achieving input has finitely many mass points, as S1 Text proves, and K is the number of mass points of the numerical solution on a grid of 801 rates.

The matched count still has one exponent per fibre, so its Fano factor hardly changes with the rate. Within each fibre the measured Fano factor falls from level to level by 0.0356 per spike on average, about nine times faster than that of the matched count, 0.0038 per spike. The matched count is therefore too regular at low counts and too noisy at high ones. A third registered test matched the Fano factor at every rate, along a line fitted to the levels of each fibre. With this K the bound held in 77.0% of the fibres, below

Table 3. Fraction of fibres with $D \leq K$ in three groups of equal size formed by the fibres' median Fano factor, for K computed with each count model. The registered tests used all 265 fibres; the split into groups was made after the tests.

fibres (Fano factor)	number	Poisson	binomial	matched	matched at every rate
most regular (0.06 to 0.31)	88	63.6%	94.3%	95.5%	87.5%
middle (0.31 to 0.41)	89	49.4%	86.5%	84.3%	77.5%
least regular (0.41 to 4.78)	88	42.0%	84.1%	67.0%	65.9%
all	265	51.7%	88.3%	82.3%	77.0%

the registered 80%, so the count model that fits the measured regularity best fails the bar. Its K is smaller than the matched K in 61 fibres: the cost of the extra noise at low rates is larger than the gain from the extra regularity at high rates. Table 3 gives the fraction for each count model in three groups of equal size, from the most regular fibres to the least regular. Every model explains the excess least well in the least regular group.

The count D has little power near a peak count of 8 when the tested levels cover only part of the fibre's range: in a synthetic check, graded fibres then met $D \leq K$ in 57.0% of cases, against 16.0% when the levels cover the whole range. A last registered test therefore replaced D by a statistic that pools all levels of a fibre. It is the smallest weighted squared distance of the level means to a monotone staircase of at most K steps, with the variance of the count matched at every rate, and it is calibrated by a parametric bootstrap. In synthetic checks it rejected graded fibres in every case and staircase fibres in at most 6%, also when the tested levels covered only part of the range. On the recorded fibres, 40.4% were consistent with at most K steps, far below the registered 80%, and 83.8% were consistent with $K + 1$ steps. Comparing the two halves of the presentations showed that the variance model of the statistic is not too small, so the statistic is not inflated. In the last registered test the spontaneous count of each fibre, measured in trials without a stimulus, set the lower end of the range on which K is computed; in 149 fibres the lowest tested level lay above it. This raised K in 55 fibres and the fraction to 49.4%, still far below 80%, with 86.8% consistent with $K + 1$ steps. By this test about half of the fibres use one rate more than the optimum.

Fig 3 shows the same result fibre by fibre.

Fig 3. Distinct mean counts against the steps of the optimal staircase. (A) For each of the 265 fibres, the number D of significantly different mean counts against the number K of steps computed for Poisson counts; points are shifted slightly so that they do not overlap, and points on or below the line $D = K$ meet the bound. (B) The same with K computed for counts matched to each fibre's Fano factor; three fibres with K above 10 are not shown, and all three meet the bound. (C) One fibre, chosen by a fixed rule among the 59 fibres that meet the bound with the matched K but not with the Poisson K: the one whose peak count is the median of that set ($A = 7.98$, spontaneous count 3.3). Mean counts per sound level with standard errors; levels that the test does not separate share a colour, so the three colours are its $D = 3$ distinct mean counts.

A larger K is easier to meet, but a registered synthetic control with counts as regular as these shows that graded tuning curves meet it in only 16.0% and 0.5% of cases at peak counts 8 and 15 (Section 4). Among the 118 fibres with a peak count of at least 8, where the test separates the two codes, the bound held in 92.4% with the matched K and in 83.9% with the K matched at every rate; this split was computed after the tests. Below that peak count a graded fibre can meet the bound as well, so those 147 fibres carry little evidence either way. Among them the bound held in 74.1% with the matched K and in 71.4% with the K matched at every rate, so they lower the fraction over all fibres and do not raise it.

Pooled over fibres, the auditory-nerve fibres thus favour the shape that Barlow's principle predicts for natural sound levels over a symmetric sigmoid, in young and in aged animals. Under Poisson counts about half of them use more distinct rates than the staircase allows. Their counts are far more regular than Poisson counts. With the staircase computed for a count matched to the mean regularity of each fibre, more than four in five fibres are consistent with it, a bound that graded tuning curves of the same regularity rarely meet. With the regularity matched at every rate, which fits the counts best, about three in four are, short of the registered 80%. Among the fibres with a peak count of at least 8, where the test can tell a staircase from

a graded curve, more than four in five are. The likelihood test, which has more power than the count D, finds about half of the fibres consistent with the optimal number of steps and nearly nine in ten consistent with one step more. The fibres use few rates, close to the optimal number; about half need more steps than the optimum, most of them one more.

3 Discussion

For a neuron whose spike count in a window is Poisson, Barlow's question has an exact answer. The most informative rate map is a staircase, its steps sit at the quantiles of the stimulus distribution, and its values are the mass points of the capacity-achieving input of the counting channel (Proposition 1). The stimulus distribution decides only where the steps sit. How many steps there are depends on the peak count and the spontaneous count alone, and Theorem 3 proves that the number grows at least like $0.056\,A^{2/3}$, faster than the $\sqrt{A}$ rates that a count of resolvable levels would allow.

Laughlin's rule is the limit of this answer when noise is negligible: the response follows the distribution function of the stimulus. With Poisson noise the equalization takes place in the square root of the rate, and the staircase approaches that curve as the peak count grows. The computations of Nikitin and colleagues found staircases numerically [4]; the proposition gives them for every continuous stimulus law, and the theorem fixes how fast their number grows. Kostal and Shinomoto found two levels for a neuron observed in continuous time [10]; counting in a window is what makes the number of levels grow.

Pooled over fibres, the auditory-nerve fibres favour the predicted shape, and they use few rates, close to the optimal number. Of the registered tests on the fibres, the tests of shape held, the tests of discreteness under Poisson counts failed, and the test that the counts are as variable as Poisson counts failed. Of the five tests of discreteness with regular counts, the two that count significant differences held and the three others failed, including both likelihood tests, which have the most power. Their rate-level functions favour the asymmetric shape predicted for a bell-shaped distribution of sound levels over the symmetric sigmoid, in young and in aged animals, and over a curve of the opposite asymmetry. A curve of the same direction with a stronger asymmetry fits them better, so the data support the direction of the asymmetry more than the exact form. Under Poisson counts about half of them show more distinct rates than the staircase allows, mostly one more, and neither drift across presentations nor a longer window explains the excess. Their counts, however, have about a third of the variance of Poisson counts with the same mean, and a less noisy count supports more rates. For a count matched to the mean dispersion of each fibre, 82.3% of the fibres stay within the staircase, and 88.3% do for the most regular binomial count with the same range. For a count whose dispersion follows the measured fall of the Fano factor with the rate, 77.0% do, below the registered 80%. The agreement is weakest in the least regular third of the fibres, where 65.9% to 84.1% stay within it, depending on the count model. The number of steps depends on the 50 ms window through the peak count and on the spontaneous count. The distribution of sound levels sets only where the steps sit. For analyses of efficient coding with spike counts, the noise model decides the predicted number of rates. Under the Poisson model about half of the fibres use more rates than the optimum. Under models with their measured regularity most of them are consistent with the optimal staircase, from 77.0% to 88.3% of the fibres depending on how the regularity is modelled. A likelihood test with more power, on the range from each fibre's spontaneous count, finds 49.4% consistent with the optimal number of steps and 86.8% with one step more. The fibres thus use few rates; about half need more than the optimum, most of them one more.

The regularity of the counts has a known physiological source. An auditory-nerve fibre cannot fire again within an absolute refractory period of about 0.6 ms [7]. Refractoriness makes the spike train of a neuron firing near its maximal rate more regular than a Poisson process with the same rate [16]. The binomial count of the tests above, at most one spike in each short bin, is the simplest model of this effect. The theory treats one fibre and its count in one window, and three things lie outside it. First, each sound level is coded by many fibres. Dynamic range adaptation is found in fibres of all spontaneous rates, and it improves the precision of level coding by the fibre population [8]. Fibres whose steps sit at different sound levels can together resolve more levels than any one of them; whether the population code is optimal is not tested here. Second, the count discards spike timing, which a reader of the full spike train would use, as in the continuous-time result above. Third, spikes cost energy. Under an energy budget at a peak count of 3, a

Table 4. Notation of the main text.

symbol	meaning
A	peak count: the largest expected count in the window above the spontaneous count
λ	spontaneous count: the expected count in the window without a stimulus
r	expected count in the window at a given stimulus; the rate times the window
K	number of steps of the optimal staircase, the number of mass points of the capacity-achieving input
D	number of significantly different mean counts across the tested sound levels of a fibre
C, $i(x)$	capacity of the counting channel; relative entropy between the count distribution at rate x and the optimal output law
L	sound level, in dB
μ, σ	mean and spread of the Gaussian distribution of sound levels
σ_{s}	scale of the symmetric sigmoid
r_0	spontaneous count of a fitted shape
d	log-likelihood of the predicted shape minus that of the sigmoid, for one fibre
F	Fano factor: the variance of a count divided by its mean
ζ	exponent of the counts of the matched test; the value 1 gives Poisson counts
n	number of trials of the binomial count
m_l, n_l	mean count and number of presentations at sound level l
T	statistic of the likelihood test

graded code gains over the best on-off code at most 0.8% of the capacity at that budget (S1 Text).

Two limits of the present tests are stated where they arise. The shape test used the form of the optimal map for a small spontaneous count. The exact form for any spontaneous count, given in Section 2.1, was tested as well and describes the fibres worse, as if the spontaneous count were an offset outside the range over which the information is optimized; why this is so is open. The conjectured law for the exact number of steps held to within 0.15 in the peak count up to 47 steps and failed at 67. There it is applied below the range on which its chain fit was made, and three further predictions of it are recorded and not yet computed. This failure does not touch the tests on fibres: they use the number of steps computed directly for each fibre, at peak counts far below those where the law failed. On six further public datasets from other species and areas, whose counts are close to Poisson counts, an earlier test of a different design did not find discrete levels (S1 Text). That test lies outside the conditions of the theory in two further ways. The window in which cortical neurons are read is not known, and the number of steps grows with the window. The stimuli were a few conditions chosen by the experimenter, and three of the datasets come from motor and frontal areas, which do not encode a stimulus distribution in Barlow's sense.

4 Materials and Methods

Table 4 lists the symbols of the main text. Appendix A uses further symbols, defined where they first appear and used only there.

4.1 Recordings

The recordings are the public single-unit data of Heeringa and colleagues [7], collected from 104 Mongolian gerbils (*Meriones unguiculatus*) of either sex. The original experimental procedures were in accordance with the ethics authorities of Lower Saxony, Germany, under the permit numbers given in [7]; no new animal experiment was done for this study. In the rate-level recordings, tones at the best frequency were played at up to 27 sound levels with 10 presentations each. In the phase-locking recordings, tones were played at 5 to 8 levels with 20 to 40 presentations each. Spikes were counted from tone onset in a window of 50 ms, and the silent condition was excluded. The young-adult group holds animals below 12 months of age and the quiet-aged group animals above 36 months, as in Table 1 of Heeringa and colleagues [7].

4.2 Shape of the rate-level function

A rate-level recording entered the test if it had at least 8 levels, at least 5 presentations at every level, and a range of level means of at least one spike per window. In all, 950 recordings passed, each from a different fibre. The predicted shape $r_0 + A\,\Phi((L-\mu)/\sigma)^2$ and the symmetric sigmoid $r_0 + A/(1+e^{-(L-\mu)/\sigma_s})$ were fitted to the single-trial counts by Poisson maximum likelihood. Both shapes have four parameters, and each fit took the best of 12 fixed starting points of a bounded quasi-Newton search, a standard optimizer that keeps every parameter within its bounds. The bounds were r_0 between 0 and 50, A between 0 and 100, μ between -20 and 120 dB, and σ and σ_s between 0.5 and 60 dB. For each fibre d is the log-likelihood of the predicted shape minus that of the sigmoid. The test passed if the bootstrap 95% lower bound of the mean of d over fibres was above zero and the median fitted σ lay between 6 and 20 dB; the bootstrap drew the fibres again with replacement 2000 times. Each synthetic fibre of the controls had 27 sound levels from 0 to 78 dB in steps of 3 dB, with 10 presentations each. Its counts were Poisson, with a mean that followed one of the two shapes: spontaneous count 0.5, peak count 8, midpoint 40 dB, and width 12 dB for the predicted shape or scale 5 dB for the sigmoid. The rule passed on fibres drawn from the predicted shape and failed on fibres drawn from the sigmoid (Table 5). The replication in the two age groups used the same values of d and the same rule for the lower bound.

4.3 Number of distinct rates

A phase-locking recording entered the test if it had at least 4 levels, at least 20 presentations at every level, and a range of level means of at least one spike per window. For each fibre, the spontaneous count λ is the mean count at the lowest level and the peak count A is the range of the level means. The number K of mass points of the capacity-achieving input on $[\lambda, \lambda + A]$ was found by solving the capacity problem numerically for the fibre's λ and A: rates are added one at a time where the optimality condition fails, until it holds everywhere. In words, the condition says that no rate left out of the input would make the count more informative than the rates already in it. Close to a peak count where a new rate appears, this numerical solution can hold the new rate with a weight below 0.005. Two of the 265 fibres have such a rate, and counting it decides their outcome, so the Poisson fraction below is uncertain by at most two fibres. The number D of distinct rates is one plus the number of neighbouring sound-level pairs whose counts differ by the exact conditional binomial test, with Holm's correction at level 0.05 within the fibre. Given the total count of two levels, the share of the first is binomial when their mean counts are equal, which is what the test checks. For example, a fibre whose mean counts at three sound levels are 5, 5 and 12, where only the second and third levels differ significantly, has $D = 2$: the first two levels share one rate and the third has another. The test passed if $D \le K$ in at least 80% of fibres. This bar lies below the fractions of 93% to 100% that synthetic fibres with a staircase reached in the controls, and far above the fraction of 0% that graded fibres reached at peak counts 8 and 15. The controls used synthetic fibres with 7 sound levels and 40 presentations at peak and spontaneous counts $(A, \lambda) = (3, 0.5)$, $(8, 0.5)$ and $(15, 1)$ (Table 5). The test therefore separates the two codes only from A of about 8 upward. Of the 265 fibres, 118 have A of at least 8. With Poisson counts $D \le K$ held in 61 of them and in 76 of the 147 others, both 51.7%; this split was computed after the test and is not part of its registered rule. A fibre counted as drifting if, at some level, the mean count of the first half of the presentations differed from that of the second half by more than the two-sided Bonferroni threshold at 0.05 for its number of levels. The drift test passed if at least 30% of fibres drifted. Its controls are in Table 5. The longer window of 75 ms is the longest that lies inside every trial.

4.4 Dispersion of the counts and the binomial channel

For every fibre and sound level with a mean count of at least one, the Fano factor is the sample variance of the counts across presentations divided by their mean. The Fano factor F of a fibre is its median over these levels. The registered rule asked whether the median over all fibres and sound levels lay between 0.8 and 1.2. On 265 synthetic fibres with 6 sound levels and 30 presentations, the rule passed for Poisson counts and failed for negative binomial and binomial counts (Table 5).

For the binomial channel the count at rate r is binomial with n trials and success probability r/n. Its K was found by the same method as for Poisson counts, on a grid of 801 rates in $[\lambda, \lambda + A]$, and mass points

closer than 0.2 in the coordinate $2\sqrt{r}$ were merged. On a grid of rates the numerical solution can split one mass point into two neighbouring grid points just after a new rate appears, 0.05 to 0.07 apart in this coordinate in the cases examined, where the noise has unit width; merging removes these splits. It can only lower K, which makes the bound $D \le K$ harder to meet. The registered rule took for n the smallest integer $n \ge \lambda + A$ for which the Fano factor $1 - \bar{m}/n$ at the mean $\bar{m}$ of the level means does not exceed F. Because $1 - \bar{m}/n$ grows with n, this integer is $\lceil \lambda + A \rceil$ whenever it exists. In 102 fibres no such integer exists, and in the first run the code then fell back to Poisson counts, against the purpose of the rule; $D \le K$ held in 72.5% of fibres in that run. Before the second run the rule was changed to use $n = \lceil \lambda + A \rceil$ in these fibres as well, and the change was recorded in the log with the outcome of the first run. The bar, the data and D were unchanged, and the change raised the fraction. The first run counts as void, and the reported result, 88.3%, is that of the second run. A binomial count is never more variable than a Poisson count with the same mean, so it cannot match a Fano factor of 1 or more; the 5 fibres with $F \ge 1$ therefore keep Poisson counts. In synthetic checks before the data were used, the binomial K at $n = 4000$ equalled the Poisson K at $(A, \lambda) = (3, 0.5)$, $(8, 0.5)$ and $(15, 1)$, and it never decreased as n decreased, rising by one step at $n = \lceil \lambda + A \rceil + 1$ for $(8, 0.5)$ and $(15, 1)$. The test of D assumes Poisson counts; for counts less variable than Poisson counts it finds fewer differences, so D is not inflated by the dispersion. This was checked on 1000 synthetic fibres with one mean count at all 7 sound levels and 40 presentations (Table 5).

In the matched test the count at rate r has probabilities proportional to $e^{ty}/(y!)^{\zeta}$, with t chosen so that the mean is r, and the count is zero at $r = 0$. The value 1 of ζ gives Poisson counts, and values above 1 give less variable ones. The distribution was truncated where the last retained count had probability below 10^{-12}. For each fibre ζ is the root in $[1, 50]$ of the condition that the Fano factor at $\bar{m}$ equals F, found numerically to within 10^{-12}. A fibre whose Fano factor stayed above F at the upper end of this interval was to keep the value 50 and be reported apart from the others; there was none. The quartiles of ζ over the fibres were 2.33, 2.99 and 3.60; at a mean count of 3 the median value gives a Fano factor of 0.374. Values of ζ of at least 1 never give counts more variable than Poisson counts, so the 5 fibres with $F \ge 1$ keep the value 1. In synthetic checks before the data were used, this K equalled the Poisson K at the value 1 of ζ for the three pairs above. It never decreased as ζ increased: for ζ equal to 1, 1.5, 2, 3 and 5 it was 3, 3, 4, 4 and 6 at $A = 8$ and 4, 4, 5, 5 and 7 at $A = 15$.

In the test matched at every rate, a least-squares line through the Fano factors of the levels with a mean count of at least one gives each fibre a Fano factor at every rate, kept between 0.05 and 1. At each rate the count is of the matched kind, with the value of ζ between 1 and 50 that gives the Fano factor of that line, found on 41 rates and interpolated to the grid. The quartiles of the fitted slope over the fibres were -0.0573, -0.0355 and -0.0182 per spike. In synthetic checks before the data were used, this channel reproduced the K above when ζ was held fixed. Its graded control, at the median line of the fibres, is in Table 5.

The likelihood test uses, for each fibre, the level means m_l over n_l presentations and the variance $V(m)$ of the count matched at every rate. Its statistic T is the smallest value of $\sum_l n_l (m_l - \mu_l)^2 / V(\mu_l)$ over all monotone maps that take at most K values on contiguous groups of levels. Here μ_l is the value of the map at level l, and each value is the mean of its group, weighted by the numbers of presentations. Its p-value comes from 200 sets of counts drawn from the matched count at the fitted group means, and a fibre counts as consistent with at most K steps if $p \ge 0.05$. The controls drew synthetic fibres from the median line, with K computed from the observed range of each, as on the data (Table 5). Splitting the presentations of each level into odd and even ones, the squared difference of the means of the two halves averaged 1.008 times its value under the variance model over all 1549 levels of the 265 fibres, so the model is not too small.

A larger K is easier to meet, so the graded control was repeated for regular counts, registered before it was run. Synthetic fibres with 7 sound levels, 40 presentations and Conway-Maxwell-Poisson counts with ζ equal to 3, close to the median of the fibres, were scored with the K of these counts. This K is 4 at $(A, \lambda) = (8, 0.5)$ and 5 at $(15, 1)$, and the outcomes are in Table 5. Among the 118 recorded fibres with A of at least 8, $D \le K$ held in 92.4% with the matched K, a fraction computed after the test.

4.5 Count changes of the Poisson channel

A count change A_K was located by following the optimal input with $K - 1$ mass points upward in A. At each step the information was maximized for the current set of rates by Newton's method, with each step

Table 5. Synthetic controls of the registered tests, each run before that test was applied to the recorded fibres. Each row gives the synthetic fibres, what the rule or statistic must do on them, and what it did. (A, λ): peak and spontaneous count.

test	synthetic fibres	required	outcome
shape	200 from the predicted shape	pass	LB 0.0802, passed
shape	200 from the symmetric sigmoid	fail	LB -0.2709, failed
$D \le K$, Poisson	staircase, $(A, \lambda) = (3, 0.5)$, $(8, 0.5)$, $(15, 1)$	meet	0.97, 1.00, 0.93
$D \le K$, Poisson	graded, same three pairs	not meet	met in 0.70, 0.00, 0.00
drift	stationary	not marked	marked in 0.045
drift	rate up by 25% in the second half	marked	marked in 0.818
Fano factor in $[0.8, 1.2]$	Poisson counts	pass	median 0.965, passed
Fano factor in $[0.8, 1.2]$	negative binomial, Fano factor 1.6	fail	median 1.535, failed
Fano factor in $[0.8, 1.2]$	binomial, Fano factor 0.6	fail	median 0.571, failed
D on one mean count	Poisson, mean counts 3 and 8	$D = 1$	$D > 1$ in 3.2% and 4.0%
D on one mean count	matched counts, $\zeta = 3$	$D = 1$	$D > 1$ in none
likelihood test	staircase, median line, $(8, 0.5)$ and $(15, 1)$, full and partial range	reject at most 10%	rejected 3%, 3%, 5%, 6%
likelihood test	graded, same four cases	reject at least 70%	rejected 100% in all four
$D \le K$, matched	staircase and graded, $\zeta = 3$, $(8, 0.5)$ and $(15, 1)$	meet / not meet	staircase 100%, 100%; graded 16.0%, 0.5%
$D \le K$, matched at every rate	staircase and graded, median line, $(8, 0.5)$ and $(15, 1)$	meet / not meet	staircase 100%, 100%; graded 1.0%, none

limited in size, in arithmetic of 160 bits (about 48 digits). The optimality condition compares, at each rate x, the relative entropy $i(x)$ between the count distribution at x and the count distribution of the optimal input; it holds when $i(x)$ is nowhere larger than at the rates in use. The largest value of $i(x) - i(0)$ away from the rates in use was evaluated with 40 significant digits. The first change of sign of this value, located by repeated linear interpolation within an interval below 2×10^{-6}, is the peak count at which the optimal input gains a new rate. In words, a new rate appears at the peak count where some rate outside the input first becomes as informative as the rates in use. On the branch with 23 rates this procedure placed A_{24}, the peak count at which the 24th rate appears, at 385.381131, against 385.3810, the value of Table 1. That value came from an earlier, independent method: the new rate was inserted just above the count change with a small weight and followed downward in A while its weight decreased, and the count change is the peak count at which that weight reaches zero. This zero was found by linear extrapolation from weights 5×10^{-4} and 2.5×10^{-4}. A first walk with the method used here gave 385.381193. The predictions of Table 1 are scored against 385.3810; at the later values their misses change by at most 2×10^{-4}, far inside the tolerance of 0.15.

4.6 Proved counts

A computed optimal input is a numerical point, and a numerical point can be wrong without any sign of it. A count is proved instead by ball arithmetic, the interval arithmetic with midpoint-radius balls: every number is carried as a midpoint and a radius that contains the exact value, so that every inequality it confirms holds for the exact quantities. The proof has three parts. First, a test of Krawczyk's type shows that the equations of a stationary input with the given number of rates have exactly one solution in a small box around the computed point. Second, every weight in that box is positive, so the solution is a probability distribution with that number of rates. Third, the relative entropy $i(x)$ of this input stays at or below the capacity on all of $[0, A]$. Near each rate this follows from a negative second derivative on an enclosing interval, and between the rates from Taylor bounds on pieces that are split until each piece is decided. The capacity-achieving input is unique and the optimality condition is sufficient [3], so the input found is the

Table 6. The claims of the paper and their evidence. Proved: a proof in the paper or in S1 Text. Proved by computer: a computer-assisted proof (Section 4.6). Registered: a test whose rule was fixed in the dated log before its data were analysed; dated: its date is confirmed by an independent authority. After the tests: a reading computed after a registered test, not part of its rule.

claim	where	evidence
optimal tuning curve is the quantile staircase	Proposition 1	proved
number of steps at least $0.056\,A^{2/3}$ for large A, any fixed spontaneous count	Theorem 3, Appendix A	proved; one step checked in interval arithmetic
finitely many rates for the matched counts	S1 Text	proved
exact number of steps follows the conjectured law	Section 2.3	conjectured; predictions registered, six of seven held
24 rates at $A = 412.42711264916346$	Section 2.3	proved by computer
count changes at 21 to 24 steps predicted before computation	Table 1	registered on 9 September 2026, log only
count changes at 25, 47 and 67 steps predicted before computation	Table 1	registered, dated
shape of the rate-level functions, in both age groups	Section 2.4	registered, dated; held
distinct rates within the staircase, seven forms of the test	Section 2.4	registered, dated; two held, five failed
counts as variable as Poisson counts	Section 2.4	registered, dated; failed
exact square-root form of the shape, against the form tested first	Section 2.4	registered, dated; failed
predicted direction of asymmetry, against the opposite direction	Section 2.4	registered, dated; held
predicted form against a more skewed curve of the same direction	Section 2.4	after the tests; the other curve fits better
discrete levels in six earlier datasets from other species and areas	Section 3, S1 Text	registered, log only; not found
fractions for $A \geq 8$ and by groups of Fano factor	Section 2.4	after the tests

optimal one and the count is exact. The computation truncates the counts at a large value; a separate bound on the omitted tail shows that it is far below every margin of the proof. At $A = 412.42711264916346$ the smallest margin was 1.24×10^{-25} and the tail bound 4.4×10^{-46}.

4.7 Registration

Every test of Section 2.4 and every prediction of Table 1 was recorded in a dated log before the data were analysed or the count change was computed. Entries from 14 October 2026 on include every test of Section 2.4 and the predictions for 47 and 67 steps. They carry a certificate of their date from an independent authority: a dated signature from a public dating service and an OpenTimestamps proof, both of which anyone can check without trusting the author. The predictions for 21 to 25 steps were entered on 9 September 2026. For 25 steps the copy of the log dated on 14 October by that authority came before the computation, made on 15 October; for 21 to 24 steps the order of prediction and computation rests on the log alone. The log is deposited with the code named under Data availability. Each entry holds the decision rule, the readout script and the outcome of the readout on synthetic data. The outcome of each test is the last line printed by its readout script, and failed tests are reported with the passed ones. Table 6 lists each claim of the paper with the kind of evidence behind it.

Data availability

The recordings are the public dataset 001262 of the Distributed Archives for Neurophysiology Data Integration (DANDI) (`https://dandiarchive.org/dandiset/001262`), described in [7]. The code that selects the fibres, fits the rate-level functions and computes every number of Section 2.4 and Section 2.3 is in the repository `https://github.com/Tristan-tech-ai/how-many-firing-rates`, together with the dated registration log of the tests and the files that certify its dates.

Competing interests

The author has declared that no competing interests exist.

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Use of AI. Claude (Anthropic) was used as an assisting tool during the preparation of this manuscript, primarily for computational support and language editing. The research questions, mathematical ideas, methodology, interpretation of results, and overall direction of the work were developed by the author. All computational results and certificates reported in the manuscript were recomputed and verified by the author, and the cited works were checked by the author. The author is responsible for the content and conclusions of this work.

A Proof of Theorem 3

In this appendix r denotes a reference output law, not the tuning curve of Section 2.1. A λ with a subscript is a constant of the proof; λ without one is the scale parameter of H_λ and S_λ in Proposition 13 and Lemma 14, and the spontaneous count in Section A.8. Here $D(x)$ is the relative entropy, or Kullback–Leibler divergence, between the count distribution at rate x and the output distribution of the optimal input; it is unrelated to the number D of distinct rates of Section 2.4.

Plan of the proof. The proof has four steps, one for each of the subsections that follow. (1) It compares the optimal input π with an explicit continuous input ρ_∞, restricted to $[0, A]$ as ρ_A, whose output r is known in closed form (Section A.3). (2) The optimality conditions of π bound the ratio of its output m to r from above and below over the whole range (Section A.4). (3) A weighted identity then shows that m is exponentially close to r in the bulk of the range (Proposition 13). (4) Finally, an input whose output is that close to r on a window must have many mass points in the window, because trigonometric test functions of high order would otherwise separate the two outputs (Lemma 17). Summing over disjoint windows gives the bound of Theorem 3. Section A.8 extends the proof to a spontaneous count.

A.1 Setting and facts from the literature

Until Section A.8 the spontaneous count is $\lambda = 0$; that section gives the changes for $\lambda > 0$. For $y \in \mathbb{N}_0 = \{0, 1, 2, \dots\}$ put $p(y \,|\, x) = e^{-x} x^y / y!$; P_x and E_x denote probability and expectation for $Y \sim \mathrm{Poisson}(x)$. Let $\pi = \sum_{j=1}^{K} w_j \delta_{x_j}$, with weights w_j, be the capacity-achieving input, $m = \mathsf{P}\pi$ its output law (P is the Poisson kernel), C the capacity in nats and $D(x) = D(p(\cdot \,|\, x) \,\|\, m)$.

(F1) π exists, is unique and has finitely many mass points [3]; it is optimal if and only if $D(x) \le C$ on $[0, A]$, with equality on its support [12, Lemma 1].

(F2) 0 and A are mass points of π [17, Corollaries 2 and 4].

(F3) $E[X \,|\, Y = y] = (y+1)m(y+1)/m(y)$ and $\frac{d}{dx} E[f(Y) \,|\, X = x] = E[f(Y+1) - f(Y) \,|\, X = x]$ for $x > 0$ [12, Appendix D]. Since $p(y_0 \,|\, x) = p(y_0 - 1 \,|\, x)\, x / y_0$, the posterior of X given $Y = y_0$ is the posterior given $Y = y_0 - 1$ tilted by x.

(F4) $E[X \mid Y = y]$ is nondecreasing in y [12, Lemma 3]. With (F3) this is the convexity of $G(y) = \log(y!\, m(y))$, whose increments are $\log E[X \mid Y = y]$.

A.2 A reference output and the exact form of the optimality conditions

Let $\varphi(y) = y\psi(y+1) - y$, so that $\varphi(y+1) - \varphi(y) = \psi(y+1)$, and put

$$r(y) = \frac{e^{-(C+1)+\varphi(y)}}{y!}, \qquad c_* = r(0) = e^{-(C+1)}, \qquad g = \log(m/r).$$

The function r is a reference output law, compared with m below.

Lemma 4. *For $x \geq 0$ one has $E_x\psi(Y+1) = \log x + E_1(x)$ and $E_x\varphi(Y) = x\log x - x + 1 - \delta_0(x)$, where $\delta_0(x) = e^{-x} - xE_1(x)$ satisfies $0 < \delta_0(x) \leq e^{-x}/(x+1)$ and $\delta_0(0) = 1$.*

Proof. With Euler's constant γ_E one has $\psi(y+1) = -\gamma_E + H_y$ with $H_y = \int_0^1 (1-(1-s)^y)\, s^{-1} ds$, so $E_x H_Y = \int_0^1 (1 - e^{-xs}) s^{-1} ds = \gamma_E + \log x + E_1(x)$. By (F3), $\frac{d}{dx} E_x\varphi(Y) = E_x\psi(Y+1)$, and $E_0\varphi(Y) = 0$; integrating gives $\int_0^x (\log s + E_1(s))\, ds = x\log x - x + xE_1(x) - e^{-x} + 1$. The bounds follow from $e^{-x}/(x+1) < E_1(x) < e^{-x}/x$. □

Proposition 5. *For every $x \geq 0$ one has $D(p(\cdot \mid x) \,\|\, r) = C + \delta_0(x)$. Hence $C - D(x) = E_x g(Y) - \delta_0(x)$, and the optimality conditions read $E_x g(Y) \geq \delta_0(x)$ on $[0, A]$, with equality at every mass point. We also have $g(0) = 1$.*

Proof. The log ratio is $\log p(y \mid x) - \log r(y) = -x + y\log x + C + 1 - \varphi(y)$; take E_x and use Lemma 4 with $E_x Y = x$. Since 0 is a mass point (F2) and $p(\cdot \mid 0) = \delta_0$, $C = D(0) = -\log m(0)$, so $m(0) = e^{-C} = e\, r(0)$. □

For $z = m/r > 0$ one has $(z-1)\log z = \mathcal{L}(1, z)(\log z)^2$ and $(z-1)^2 = \mathcal{L}(1, z)(z-1)\log z$; so g has the sign of $m - r$ and $(m-r)g \geq 0$ pointwise.

Proposition 6 (weighted identity)**.** *For every bounded measurable $\chi \geq 0$ on $[0, A]$ and every finite measure $\rho \geq 0$ on $[0, A]$, with $\mu = \rho - \pi$,*

$$\mathcal{W}(\chi) := \int \chi\,(C - D)\, d\rho \;\geq 0, \qquad \mathcal{W}(\chi) = \sum_{y \geq 0} g(y)\, \mathsf{P}(\chi\mu)(y) - \int \chi\, \delta_0\, d\mu.$$

Proof. One has $\mathcal{W} \geq 0$ by the optimality inequality, and $\int \chi(C - D)\, d\pi = 0$ by the equality on the support. Insert Proposition 5 and exchange sum and integral; the exchange is justified because $|g(y)|$ grows at most quadratically in $y - \lfloor A \rfloor$ beyond A (Lemma 15(iii)), while $\mathsf{P}(\chi|\mu|)(y)$ decays like $p(y \mid A)$ for $y > A$. □

A.3 The reference input

Lemma 7. *(i) Let $\pi_Z(u) = (ue^u - e^u + 1)/(u(e^u - 1)^2) > 0$ on $(0, \infty)$, let $S \geq 0$ be the driftless subordinator variable (the value at time one of an increasing Lévy process without drift; only this random variable is used, never the process) with Laplace exponent $\int (1 - e^{-pu})\pi_Z(u)\, du$, that is, $E\, e^{-pS} = \exp\bigl(-\int (1 - e^{-pu})\pi_Z(u)\, du\bigr)$, and put $Z = e^{-S}$. Then $EZ^p = \exp(p\psi(p+1) - p - \log\Gamma(p+1))$ for $p \geq 0$, so $r(y) = c_*\, EZ^y$. (ii) $\rho_\infty(dx) = c_*\, E[Z^{-1} e^{-x(1-Z)/Z}]\, dx$ is a completely monotone density on $(0, \infty)$ with $\mathsf{P}\rho_\infty = r$. (iii) $\hat{\rho} := \rho_\infty / c_*$ satisfies $\hat{\rho}(x) \leq x^{-1/2}$ for $x \geq 1$. (iv) The restriction ρ_A of ρ_∞ to $[0, A]$ has mass $m_A \leq \exp(-\int \delta_0\, d\rho_A / m_A) < 1$.*

Proof. (i) $k(p) := p\psi(p+1) - p - \log\Gamma(p+1)$ has $k(0) = 0$ and $k'(p) = p\psi'(p+1) - 1$. From $\psi'(p+1) = \int_0^\infty ue^{-pu}(e^u - 1)^{-1} du$, integration by parts gives $p\psi'(p+1) = 1 + \int e^{-pu} \frac{d}{du}[u/(e^u - 1)]\, du$, and $\frac{d}{du}[u/(e^u - 1)] = -u\pi_Z(u)$. Hence $k(p) = -\int (1 - e^{-pu})\pi_Z(u)\, du$; since $\int \min(1, u)\pi_Z < \infty$, this is the Laplace exponent of a driftless subordinator. We also have $\pi_Z > 0$ because $ue^u - e^u + 1 > 0$ for $u > 0$. (ii) The density $z^{-1}e^{-x(1-z)/z}$ on $(0, \infty)$ has Poisson output z^y, because $\int_0^\infty e^{-x/z} x^y / y!\, dx = z^{y+1}$; mix over $z = Z$ (positive terms, Fubini). (iii) $\pi_Z(u) \geq e^{-u}/(2u)$: with $v = e^u$ this is $2v^2 \log v - 3v^2 + 4v - 1 \geq 0$ for

$v \ge 1$, which vanishes at $v = 1$ with zero derivative and has second derivative $4 \log v \ge 0$. The right side is the Lévy density of the Gamma$(1/2, 1)$ subordinator, so $S = S_1 + S'$ with S_1 a Gamma$(1/2, 1)$ variable and $S' \ge 0$ independent of S_1. For $x \ge 1$, $s \mapsto e^{s - x(e^s - 1)}$ is decreasing, and $\hat\rho(x) = E\, e^{S - x(e^S - 1)} \le E\, e^{S_1 - x(e^{S_1} - 1)} \le E\, e^{-(x-1)S_1} = x^{-1/2}$. (iv) $\bar\rho = \rho_A / m_A$ is a feasible input with output R_A/m_A, where $R_A = \mathsf{P}\rho_A \le r$. By Proposition 5, $I(\bar\rho) = C + \int \delta_0\, d\bar\rho + \log m_A - \sum (R_A/m_A) \log(R_A/r) \ge C + \int \delta_0\, d\bar\rho + \log m_A$, and $I(\bar\rho) \le C$. □

Lemma 8 (two-sided sizes)**.** *(i) $\hat\rho(x) \ge c_1/(x \log x)^{1/2}$ for $x \ge x_0$, with absolute constants c_1, x_0. (ii) Under Proposition 11, $c_* \le C_1 A^{-1/2}$; with Lemma 15(vii), $c_* = \Theta(A^{-1/2})$.*

Proof. (i) $e^{s - x(e^s - 1)} \ge e^{-2xs}$ for $0 \le s \le 1$, so $\hat\rho(x) \ge e^{-2} P(S \le 1/x)$. Since $E e^{-pS} \le P(S \le s) + e^{-ps}$ and $E e^{-pS} = r(p)/c_* \ge (e/(2\pi p))^{1/2} e^{-5/(24p)}$ for $p \ge 1$ (bounds on r in Section A.4), the choice $s = 1/x$, $p = 2x \log x$ gives $P(S \le 1/x) \ge c/(x \log x)^{1/2} - x^{-2} \ge c'/(x \log x)^{1/2}$ for large x. (ii) $1 \ge \sum_{y \le A} m(y) \ge \ell \sum_{y \le A} r(y) \ge \ell c_* \sum_{1 \le y \le A} (e/(2\pi y))^{1/2} e^{-5/(24y)} \ge c'' \ell c_* A^{1/2}$. □

The right-edge remainder $T_A = r - R_A = \mathsf{P}(\rho_\infty \mathbf{1}_{(A,\infty)})$ satisfies $T_A(y) \le c_* A^{-1/2} e^{-(A-y)^2/(2A)}$ for $y \le A$.

A.4 The range of m/r

By (F4) $G(y) = \log(y!\, m(y))$ is convex, and $g = G - \varphi + C + 1$. For real $y \ge 1$,

$$\sqrt{e/(2\pi y)}\, e^{-5/(24y)} \le r(y)/c_* \le \sqrt{e/(2\pi y)}\, e^{1/(2(y+1))},$$

and r is decreasing ($r = c_* E Z^y$ with $Z < 1$); this follows from $\log(y + \tfrac12) < \psi(y+1) < \log(y+1) - 1/(2(y+1))$ and Stirling's bounds.

Lemma 9 (pointwise upper bound)**.** *For $y_0 \ge 1$ let $m_1 = E[X \mid Y = y_0 - 1]$ and $m_2 = E[X \mid Y = y_0]$. Then $g(y_0) \le 1 + y_0 \log m_1 - m_1 - \varphi(y_0) + m_1 \log(m_2/m_1)$.*

Proof. The slope $s_0 = G(y_0) - G(y_0 - 1) = \log m_1$ (F3) is a subgradient of the convex sequence G at y_0, so $g(y) \ge g(y_0) + s_0(y - y_0) - (\varphi(y) - \varphi(y_0))$ for all $y \in \mathbb{N}_0$. Take E_{x_j} at a mass point x_j: $E_{x_j} g(Y) = \delta_0(x_j)$ (Proposition 5) and $E_{x_j} \varphi(Y) = x_j \log x_j - x_j + 1 - \delta_0(x_j)$ (Lemma 4), so δ_0 cancels and $g(y_0) \le 1 + [s_0 y_0 - e^{s_0} - \varphi(y_0)] + B(x_j, e^{s_0})$ with $B(x, x') = x \log(x/x') - x + x' \ge 0$. Average over the posterior of X given $Y = y_0 - 1$, a probability on the mass points with mean $m_1 = e^{s_0}$. The posterior given $Y = y_0$ is that posterior tilted by x, so $E[B(X, m_1) \mid Y = y_0 - 1] = m_1 E[\log(X/m_1) \mid Y = y_0] \le m_1 \log(m_2/m_1)$ by Jensen's inequality. □

Proposition 10. *The function g satisfies $\sup_{y \in \mathbb{N}_0} g(y) \le \lambda_M = 1 - \psi(2) + e^{\psi(2)} = 2.1034207765\ldots$, that is, $m \le Mr$.*

Proof. One has $g \to -\infty$ because m has Poisson tails beyond A while $r(y)$ is of order $y^{-1/2}$, so g attains its maximum at some y_0. If $y_0 = 0$, $g(0) = 1$ (Proposition 5). If $y_0 \ge 1$, then $g(y_0 - 1) \le g(y_0) \ge g(y_0 + 1)$ and $\varphi(y) - \varphi(y-1) = \psi(y)$ give $m_1 \ge e^{\psi(y_0)}$ and $m_2 \le e^{\psi(y_0+1)}$. In Lemma 9, $y_0 \log m_1 - m_1 \le y_0 \log y_0 - y_0$, and $m_1 \log(m_2/m_1) \le m_1(\psi(y_0+1) - \psi(y_0)) = m_1/y_0 \le m_2/y_0 \le e^{\psi(y_0+1)}/y_0$. With $\varphi(y_0) = y_0 \psi(y_0 + 1) - y_0$,

$$g(y_0) \le f(y_0) := 1 + y_0(\log y_0 - \psi(y_0 + 1)) + e^{\psi(y_0+1)}/y_0.$$

At $y_0 = 1$ the bound is $f(1) = \lambda_M$. For $y_0 \ge 2$ the inequalities $\psi(y_0 + 1) > \log(y_0 + \tfrac12)$ and $\log(1 + u) \ge u - u^2/2$ give $y_0(\log y_0 - \psi(y_0 + 1)) \le -\tfrac12 + 1/(8y_0)$, and $\psi(y_0 + 1) < \log(y_0 + 1)$ gives $e^{\psi(y_0+1)}/y_0 < 1 + 1/y_0$; so $f(y_0) < \tfrac32 + 9/(8y_0) \le 2.0625 < \lambda_M$. □

Proposition 11 (lower bound on the input range)**.** *For every integer $y_1 \in [0, A]$ one has $g(y_1) \ge -\lambda_\ell$ with $\lambda_\ell = 2.777$, that is, $m \ge \ell r$ there.*

Proof. Fix y_1 and put $\lambda_1 = -g(y_1)$; there is nothing to show if $\lambda_1 \le 0$. For an integer $R \ge 1$ and $y = y_1 + j$ with $0 \le j \le R$, convexity of G and $g(y_1 + R) \le \lambda_M$ give $g(y) \le (1-\theta)(-\lambda_1) + \theta\lambda_M + \mathrm{def}_R(j)$ with $\theta = j/R$ and the chord defect $\mathrm{def}_R(j) = (1-\theta)\varphi(y_1) + \theta\varphi(y_1+R) - \varphi(y_1+j) \le j(R-j)/(2(y_1+1))$, by the discrete maximum principle since $\Delta^2\varphi(y) = 1/(y+1)$. Left chords ($R \le y_1$) give the same bound with $y_1 + 1$ replaced by $y_1 - R + 1$, and beyond the chords $g \le \lambda_M$. Let U_{λ_1} be the pointwise minimum of these bounds; it is nonincreasing in λ_1. The optimality inequality at the feasible point $x = y_1$ gives $E_{y_1} U_{\lambda_1} \ge E_{y_1} g \ge \delta_0(y_1) \ge 0$.

(a) $y_1 \le 3000$. For $y_1 = 0$ the claim holds because $g(0) = 1$ (Proposition 5). The bound $U_{1.8}$ is built from chord bounds, with the Poisson mass outside the chords counted at λ_M. For each integer $y_1 \in [1, 3000]$ it gives $E_{y_1} U_{1.8} - \delta_0(y_1) < 0$, with an upper bound of at most -6.67×10^{-3} (attained at $y_1 = 4$); each value is enclosed in ball arithmetic (interval arithmetic with midpoint-radius balls) at 128 bits. Hence $\lambda_1 < 1.8$ there.

(b) $y_1 > 3000$. One chord on each side with $R = \lceil c\sqrt{y_1}\,\rceil$, $c = 2.6$, and $\theta_Y = |Y - y_1|/R$ give $U \le -\lambda_1(1-\theta_Y)_+ + \lambda_M \min(1, \theta_Y) + R^2/(8(y_1 - R + 1))$. Since $E|Y - y_1| \le \sqrt{y_1}$, it follows that $E(1-\theta_Y)_+ \ge 1 - 1/c$ and $E\min(1,\theta_Y) \le 1/c$; so $\lambda_1 \le (\lambda_M/c + R^2/(8(y_1 - R + 1)))/(1 - 1/c)$. With $R \le c\sqrt{y_1} + 1$ the last term is at most $R_b(y_1) = (c^2 y_1 + 2c\sqrt{y_1} + 1)/(8(y_1 - c\sqrt{y_1}))$, which decreases on $y_1 \ge 3001$, and $R_b(3001)$ gives $\lambda_1 \le 2.77649$. □

Corollary 12. *With $z = m/r$: (LM1) $(z-1)\log z \ge \mathcal{L}(1,\ell)(\log z)^2$ on $[0, A]$, and (LM2) $(z-1)^2 \le \mathcal{L}(1, M)(z-1)\log z$ on $\mathbb{N}_0$. The value of ϱ is* 10.128.

Proof. The ratio satisfies $z \in [\ell, M]$ on $[0, A]$ and $z \le M$ on $\mathbb{N}_0$ (Propositions 10 and 11), and $\mathcal{L}(1, z)$ is increasing in z. □

A.5 Exponential decay of $m - r$ in the bulk

Weight. Fix $\xi \in (0,1)$ and an integer $\sigma \ge 1$, and put $t = 2\sigma/(\xi A)$ and $\chi(x) = (x/A)^{2\sigma} e^{t(A-x)}$. Then $\log\chi(x) = \Lambda_0 h(x/A)$ with $\Lambda_0 = tA\phi_\xi$; $\log\chi$ is concave, maximal at $x_c = \xi A$, zero at $x_L = \kappa_L A$ and at A, positive on (x_L, A) and negative on $[0, x_L)$ and beyond A, with $\chi(0) = 0$. Tangent bounds: $\log\chi(x) \le \beta_L(x - x_L)$ with $\beta_L = t(\xi/\kappa_L - 1)$, and $\log\chi(x) \le \beta_A(A - x)$ with $\beta_A = t(1-\xi)$. Let $\mathrm{Win} = \{y \in \mathbb{N}_0 : x_L \le y \le A\}$.

Objects. Put $\rho = \rho_A$ (Lemma 7), $\mu = \rho - \pi$, $\nu = \chi^{1/2}\mu$ (which has no mass at the origin since $\chi(0) = 0$), with $N := \|\nu\|_{\mathrm{TV}} \le 2e^{\Lambda_0/2}$. Put $\lambda_\pm = 1 \pm t/2$ and $H_\lambda(k) = \int p(k \,|\, \lambda x)\,\nu(dx)$; $F = \mathsf{P}\mu = r - T_A - m$, $\mathcal{P}(y) = \chi(y)^{1/2} F(y)$, $\mathcal{Q}(y) = \chi(y)^{-1/2}\mathsf{P}(\chi\mu)(y)$ and $C_m = \mathsf{P}(\chi\mu) - \chi F = \chi^{1/2}(\mathcal{Q} - \mathcal{P})$. The weighted error is $E_2 = \sum_{y \le A} \chi(y)(m - r)g \ge 0$.

Proposition 13 (decay)**.** *Fix $\xi \in (0,1)$ and $c_0 < c_0^{\max}(\xi) = \Gamma_\xi^{-2/3}$, where*

$$\Gamma_\xi = (1 + \xi\kappa_L^{-1/2})\kappa_L^{-1/4}\varrho^{1/2}/\phi_\xi.$$

Take $\sigma = \mathrm{round}(c_0 A^{1/3}\xi/(2\phi_\xi))$, so that $\Lambda_0 = c_0 A^{1/3}(1 + o(1))$, and $B^2 = \lambda_+(\Lambda_0 + 4\log A)$. There are A_0 and C_0, depending on ξ and c_0 only, such that $E_2 \le C_0 A^{1/2}$ for $A \ge A_0$.

Proof. Step 1 (identity). Proposition 6 with this χ and ρ, and $\mathsf{P}(\chi\mu) = \chi F + C_m$ on $y \le A$, give $E_2 = -\mathcal{W} - \tilde{T} + \sum_{\mathrm{Win}} g C_m + e_L + e_{\mathrm{out}} - d_0$, with $\tilde{T} = \sum_{y \le A} \chi g T_A$ and $e_L = \sum_{y < x_L} g C_m$, and with $e_{\mathrm{out}} = \sum_{y > A} g\,\mathsf{P}(\chi\mu)$ and $d_0 = \int \chi\delta_0\, d\mu$. Since $\mathcal{W} \ge 0$, the bound $E_2 \le |\tilde{T}| + |\sum_{\mathrm{Win}} g C_m| + |e_L| + |e_{\mathrm{out}}| + |d_0|$ holds.

Step 2 (exact Poisson flows). From $x^s e^{-\tau x} p(y \,|\, x) = \frac{\Gamma(y+s+1)}{\Gamma(y+1)(1+\tau)^{y+s}}\, p(y + s \,|\, (1+\tau)x)$ with $s = \pm\sigma$, $\tau = \pm t/2$: for $y \ge \sigma$ one has $\mathcal{Q}(y) = \alpha_+(y) H_{\lambda_+}(y + \sigma)$ and $\mathcal{P}(y) = \alpha_-(y) H_{\lambda_-}(y - \sigma)$, with $\alpha_+(y) = y^{-\sigma} e^{ty/2}\Gamma(y + \sigma + 1)/(\Gamma(y+1)\lambda_+^{y+\sigma})$ and $\alpha_-(y) = y^{\sigma} e^{-ty/2}\Gamma(y - \sigma + 1)/(\Gamma(y+1)\lambda_-^{y-\sigma})$; $\alpha_- \ge e^{-\sigma t/2}$. Put $\mathcal{A}_1 = \sup_{\mathrm{Win}} |\alpha_+ - \alpha_-|$ and $\mathcal{A}_2 = \sup_{\mathrm{Win}} \alpha_-$; both $\mathcal{A}_1$ and $\mathcal{A}_2 - 1$ are $O(\Lambda_0^2/A)$.

Step 3 (Cauchy–Schwarz with (LM1)). With $r_{\min} = \min_{\mathrm{Win}} r = r(\lfloor A \rfloor)$ one has $|\sum_{\mathrm{Win}} g C_m| \le (E_2/(\mathcal{L}(1,\ell) r_{\min}))^{1/2} \|\mathcal{Q} - \mathcal{P}\|_{\ell^2(\mathrm{Win})}$ and $|\tilde{T}| \le (E_2/\mathcal{L}(1,\ell))^{1/2}\tau_T$, where $\tau_T^2 = \sum_{y \le A} \chi T_A^2/r$.

Step 4 (three terms). On Win the difference is
$\mathcal{Q}-\mathcal{P}=(\alpha_+-\alpha_-)H_+(y+\sigma)+\alpha_-(H_+-H_-)(y+\sigma)+\alpha_-[H_-(y+\sigma)-H_-(y-\sigma)]$, where $H_\pm=H_{\lambda_\pm}$.
With $K_1=A+\sigma$ and the Dirichlet form $S_\lambda=\sum_k(k+1)(\Delta H_\lambda(k))^2$,

$$\|\mathcal{Q}-\mathcal{P}\|_{\mathrm{Win}}\le \mathcal{A}_1\|H_{\lambda_+}\|+\mathcal{A}_2\,t\sup_{\lambda\in[\lambda_-,\lambda_+]}\lambda^{-1}\big[((K_1+1)S_\lambda)^{1/2}+\|H_\lambda\|\big]+\mathcal{A}_2\,2\sigma\big(S_{\lambda_-}/(x_L-\sigma+1)\big)^{1/2};$$

the middle term comes from $\frac{d}{d\lambda}H_\lambda(k)=-\lambda^{-1}[(k+1)\Delta H_\lambda(k)+H_\lambda(k)]$ and Minkowski's inequality over λ, the last from writing $H(y+\sigma)-H(y-\sigma)$ as a sum of 2σ differences, each index $j\ge x_L-\sigma$ being used at most 2σ times.

Step 5 (one global bandwidth). Lemma 14 gives, with $\Delta_\lambda=1/\lambda_--1/\lambda_+$ and $h_0=\|H_{\lambda_-}\|$, for $\lambda\in[\lambda_-,\lambda_+]$: $\|H_\lambda\|\le e^{\Delta_\lambda B^2/2}h_0+Ne^{-B^2/(2\lambda_+)}$ and $S_\lambda^{1/2}\le(B/\lambda_-^{1/2})\|H_\lambda\|+Ne^{-B^2/(2\lambda_+)}(B^2/\lambda_-+1)^{1/2}$.

Step 6 (back to $\mathcal{P}$). By Step 2, $H_{\lambda_-}(k)=\mathcal{P}(k+\sigma)/\alpha_-(k+\sigma)$, so $h_0\le e^{\sigma t/2}\|\mathcal{P}\|_{\ell^2(\mathbb{N}_0)}$. On Win, (LM2) and the decrease of r give $\chi(m-r)^2\le\mathcal{L}(1,M)\,r(x_L)\,\chi(m-r)g$, so
$\|\mathcal{P}\|\le(\mathcal{L}(1,M)r(x_L)E_2)^{1/2}+P_L+\tau_T'+\|F\|_{\mathrm{out}}$ with $P_L^2=\sum_{y<x_L}\chi(m-r)^2$ and $\tau_T'^2=\sum_{y\le A}\chi T_A^2$, and with $\|F\|_{\mathrm{out}}^2=\sum_{y>A}\chi F^2$.

Step 7 (closing). Steps 4–6 give $\|\mathcal{Q}-\mathcal{P}\|_{\mathrm{Win}}\le\delta_1h_0+\rho_N$ with

$$\delta_1=e^{\Delta_\lambda B^2/2}\Big[\mathcal{A}_1+\mathcal{A}_2\frac{t}{\lambda_-}\Big(\frac{(K_1+1)^{1/2}B}{\lambda_-^{1/2}}+1\Big)+\mathcal{A}_2\frac{2\sigma B}{(\lambda_-(x_L-\sigma+1))^{1/2}}\Big],$$

$$\rho_N=Ne^{-B^2/(2\lambda_+)}\Big[\mathcal{A}_1+\mathcal{A}_2\frac{t}{\lambda_-}\Big((K_1+1)^{1/2}\Big(\frac{B}{\lambda_-^{1/2}}+\Big(\frac{B^2}{\lambda_-}+1\Big)^{1/2}\Big)+1\Big)$$
$$+\mathcal{A}_2\frac{2\sigma}{(x_L-\sigma+1)^{1/2}}\Big(\frac{B}{\lambda_-^{1/2}}+\Big(\frac{B^2}{\lambda_-}+1\Big)^{1/2}\Big)\Big].$$

With Steps 1, 3 and 6, $E_2\le\delta'E_2+b\,E_2^{1/2}+\epsilon_{\mathrm{out}}$, where $\delta'=\delta_1e^{\sigma t/2}(\mathcal{L}(1,M)r(x_L)/(\mathcal{L}(1,\ell)r_{\min}))^{1/2}$,
$b=[\delta_1e^{\sigma t/2}(P_L+\tau_T'+\|F\|_{\mathrm{out}})+\rho_N]/(\mathcal{L}(1,\ell)r_{\min})^{1/2}+\tau_T/\mathcal{L}(1,\ell)^{1/2}$ and $\epsilon_{\mathrm{out}}=|e_L|+|e_{\mathrm{out}}|+|d_0|$. If $\delta'<1$ then $E_2\le\big[(b+(b^2+4(1-\delta')\epsilon_{\mathrm{out}})^{1/2})/(2(1-\delta'))\big]^2$.

Asymptotics. One has $Ne^{-B^2/(2\lambda_+)}\le2A^{-2}$; $\mathcal{A}_1$, $\mathcal{A}_2-1$, σt, t^2A, β_A^2A, β_L^2A and $\Delta_\lambda B^2$ are $O(\Lambda_0^2/A)\to0$; $\delta_1=(1+o(1))\,tA^{1/2}B\,(1+\xi\kappa_L^{-1/2})$ with $tA^{1/2}B=c_0^{3/2}/\phi_\xi\,(1+o(1))$; and $r(x_L)/r_{\min}\to\kappa_L^{-1/2}$ by the bounds on r in Section A.4. Hence $\delta'\to c_0^{3/2}\Gamma_\xi<1$. With Lemma 15, $b=O(A^{1/4})$ and $\epsilon_{\mathrm{out}}=O(A^{1/2})$, so $E_2=O(A^{1/2})$. □

Lemma 14 (Hankel form of the ℓ^2 norms)**.** *Let J_0 and I_0 be the Bessel function and the modified Bessel function of order zero. For a finite signed measure ν on $[0,\infty)$ let $\hat\nu(s)=\int J_0(2sx^{1/2})\,\nu(dx)$. Then*

$$\sum_k p(k\,|\,\lambda x)p(k\,|\,\lambda x')=e^{-\lambda(x+x')}I_0\big(2\lambda(xx')^{1/2}\big)=\frac{2}{\lambda}\int_0^\infty e^{-s^2/\lambda}J_0(2sx^{1/2})J_0(2sx'^{1/2})\,s\,ds,$$

so that $\|H_\lambda\|^2=\frac{2}{\lambda}\int e^{-s^2/\lambda}|\hat\nu(s)|^2s\,ds$ and $S_\lambda=\frac{2}{\lambda^2}\int e^{-s^2/\lambda}|\hat\nu(s)|^2s^3\,ds$. The bounds of Step 5 of Proposition 13 follow by splitting at $s=B$ and using $|\hat\nu|\le N$ above B.

Proof. The first equality is the series of I_0. The second is Weber's second exponential integral [18, 13.31(1)], in its case of order zero, $\int_0^\infty e^{-p^2t^2}J_0(at)J_0(bt)\,t\,dt=(2p^2)^{-1}e^{-(a^2+b^2)/(4p^2)}I_0(ab/(2p^2))$ for $|\arg p|<\pi/4$, with $a=2x^{1/2}$, $b=2x'^{1/2}$ and $p^2=1/\lambda$. The norm follows by integrating against $\nu\otimes\nu$. For S_λ, compute $\frac{d}{d\lambda}(\lambda\|H_\lambda\|^2)$ on both sides: on the transform side, and as $\sum_k(k+1)(\Delta H_\lambda(k))^2$ on the output side. □

Lemma 15 (edge terms)**.** *With $\lambda_*=\max(\lambda_M,\lambda_\ell)$: (i) $|d_0|\le(1+e)c_*+2e^{\Lambda_0-x_L}$; (ii) $\tau_T^2\le c_*A^{-1/2}(1+(2\pi A)^{1/2})e^{\beta_A^2A/2}$ and $\tau_T'^2\le c_*\tau_T^2$; (iii) $|e_{\mathrm{out}}|\le2e^{\beta_A^2A/2}(\lambda_*+2\lambda_*(A^{1/2}+1)+3)$; (iv) $|e_L|\le\lambda_*\big[2+2e^{\beta_L^2A/2}+(1+M)c_*(1+2e^{1/4}(ex_L/(2\pi))^{1/2})\big]$; (v) $P_L^2\le(1+M)^2c_*^2(1+e^{1/2}(e/(2\pi))(1+\log x_L))$; (vi) $\|F\|_{\mathrm{out}}^2\le4(2\pi\lfloor A\rfloor)^{-1/2}$; (vii) $c_*\ge(1-(e/4)^A)/(M(1+2e^{1/4}(eA/\pi)^{1/2}))$, hence $r_{\min}=\Omega(1/A)$.*

Proof. (i) $\chi \le 1$ on $[0, x_L)$ and $\delta_0(x) \le e^{-x}$ there; $\int e^{-x} d\rho_\infty = \mathsf{P}\rho_\infty(0) = r(0) = c_*$ and $\sum_j w_j e^{-x_j} = m(0) = e\, c_*$ (Proposition 5); on $[x_L, A]$ one has $\chi\delta_0 \le e^{\Lambda_0 - x_L}$, and the masses of ρ_A and π are at most 1. (ii) $\chi T_A^2/r \le \chi T_A$ because $T_A \le r$; $T_A(y) \le c_* A^{-1/2} e^{-(A-y)^2/(2A)}$ and $\chi(y) \le e^{\beta_A(A-y)}$; sum over $y \le A$ by the maximum plus the integral of the unimodal summand. Finally $\tau_T'^2 = \sum \chi(T_A^2/r) r \le (\max r)\tau_T^2 = c_*\tau_T^2$, since r is decreasing with $r(0) = c_*$. (iii) For $y > A$ one has $|g(y)| \le \gamma_g(y) := \lambda_* + (y - y_A)(2\lambda_* + (y - y_A)/y_A)$ with $y_A = \lfloor A \rfloor$: the tangent of the convex G at y_A has slope $G(y_A) - G(y_A - 1) \ge \psi(y_A) - \lambda_\ell - \lambda_M$, and both $\varphi(y) - \varphi(y_A) \le (y - y_A)\psi(y)$ and $\psi(y) - \psi(y_A) \le (y - y_A)/y_A$ hold. We also have $\mathsf{P}(\chi|\mu|)(y) \le 2e^{\beta_A^2 A/2} p(y \,|\, A)$ for $y > A$, from $p(y \,|\, x) \le p(y \,|\, A) e^{-(A-x)^2/(2A)}$ for $x \le A < y$ and $\chi(x) \le e^{\beta_A(A-x)}$. Then $|e_{\mathrm{out}}| \le 2e^{\beta_A^2 A/2} E\gamma_g(Y')$ with $Y' \sim \mathrm{Poisson}(A)$, which gives (iii). (iv) $\sum_{y<x_L} \mathsf{P}(\chi|\mu|)(y) = \int \chi(x) P_x(Y < x_L)\, d(\rho + \pi)$; for $x \ge x_L$ the Poisson Chernoff bound gives $\chi(x) P_x(Y < x_L) \le e^{\beta_L(x - x_L) - (x - x_L)^2/(2x)} \le e^{\beta_L^2 A/2}$; on $y \le A$ one has $|F| \le (1 + M)r$, and r is summed with the bounds of Section A.4. (v) $P_L^2 = \sum_{y<x_L} \chi(m - r)^2 \le (1 + M)^2 \sum_{y<x_L} r^2$, with the bounds on r. (vi) For $y > A$ one has $|F| \le R_A + m \le 2p(y \,|\, A)$ because $p(y \,|\, \cdot)$ increases on $[0, y]$, and $\max_y p(y \,|\, A) \le p(y_A \,|\, y_A) \le (2\pi y_A)^{-1/2}$. (vii) $1 = \sum m \le M \sum r$, and $\sum_y r(y) \le c_*(1 + 2e^{1/4}(eA/\pi)^{1/2}) + (e/4)^A$ by the bounds on r, the last term bounding the mass beyond $2A$. □

The constants in (i)–(vii) are finite powers and exponentials of $A^{1/3}$; they enter b and ϵ_{out} as $O(A^{1/4})$ and $O(A^{1/2})$ because $\beta_A^2 A$ and $\beta_L^2 A$ are $O(\Lambda_0^2/A) \to 0$.

Corollary 16 (one window)**.** *Under Proposition 13, for $A \ge A_0$ and every set Y_W of integers inside* Win *one has $\sum_{y \in Y_W} (m - r)^2/r \le \mathcal{L}(1, M)\, C_0 A^{1/2} \exp(-\Lambda_0 \min_{Y_W} h(y/A))$, uniformly in Y_W.*

Proof. (LM2) gives $(m - r)^2/r \le \mathcal{L}(1, M)(m - r)g$ on $\mathbb{N}_0$; each term of E_2 is nonnegative and $\chi \ge \exp(\Lambda_0 \min_{Y_W} h)$ on Y_W. □

A.6 Counting mass points in a window

Test functions. For $\vartheta \in (0, 1)$ and integers $1 \le s_1 < s_2$, the window $\varpi(x) = P(s_1 \le \mathrm{Poisson}(\vartheta x) \le s_2)$ has the exact deconvolution $\Psi_\omega(y) = \sum_{s=s_1}^{s_2} \binom{y}{s} \vartheta^s (1 - \vartheta + i\omega)^{y-s}$: $E_x \Psi_\omega(Y) = e^{i\omega x} \varpi(x)$, by binomial thinning $(E_x[\binom{Y}{s} \vartheta^s (1 - \vartheta + i\omega)^{Y-s}] = e^{i\omega x} e^{-\vartheta x} (\vartheta x)^s / s!)$. We also have $|\Psi_\omega(y)| \le \Psi_0(y)(1 + \omega^2/(1 - \vartheta)^2)^{(y - s_1)_+/2}$ with $\Psi_0(y) = P(s_1 \le \mathrm{Bin}(y, \vartheta) \le s_2) \le 1$.

Lemma 17. *Let $l_1 = s_1/\vartheta$, $l_2 = s_2/\vartheta$, $\omega_0 = 2\pi/(l_2 - l_1)$ and $W = [a_1, a_2]$ with $a_1 \le l_1 < l_2 \le a_2 \le A$. Let k be the number of mass points of π in W and let $Y_W = [b_1, b_2]$ be a set of integers. Put $\epsilon_W = \max(\varpi(a_1), \varpi(a_2))$, $\gamma = (l_2 - l_1) \min_{[l_1, l_2]}(\varpi c_* \hat{\rho})$, $\Omega_n = n\omega_0$ and $g_n(y) = (1 + \Omega_n^2/(1 - \vartheta)^2)^{(y - s_1)_+/2}$; put also $\tau_n = M \sum_{y \notin Y_W} g_n(y) \Psi_0(y) r(y)$, and let $\eta_W^2 = \sum_{Y_W} (R_A - m)^2/r$ and $r_W = \sum_{Y_W} r$. Fix $r_b > 1$, put $n(k) = \lceil r_b k/(r_b - 1) \rceil - 1$, and let $n_* \ge 1$ be an integer with*

$$(a)\ \ r_b \epsilon_W/\gamma \le 1/4, \qquad (b)\ \ (r_b(2n_* + 1))^{1/2} \tau_{n_*}/\gamma \le 1/4.$$

If $k_C := \frac{r_b - 1}{r_b} \frac{(1 - \vartheta)(l_2 - l_1)}{2\pi} \Big(\frac{2 \log\big(\gamma/(2(r_b(2n_* + 1) r_W)^{1/2} \eta_W)\big)}{b_2 - s_1} \Big)^{1/2} > 0$ and $n(\lceil k_C \rceil - 1) \le n_*$, then $k \ge k_C$.

Proof. Suppose $k < k_C$ and put $n = n(k)$ ($n = 0$ if $k = 0$); then $n \le n(\lceil k_C \rceil - 1) \le n_*$, so (b) holds at n since τ_n increases with n. Let V be the space of trigonometric polynomials $\sum_{l=0}^n c_l e^{il\omega_0 x}$ vanishing at the k mass points in W; $\dim V \ge n + 1 - k$, so $(n + 1)/\dim V \le r_b$. Take an orthonormal basis (P_i) of V in $L^2(\mu_\varpi)$, where $\mu_\varpi = \varpi \rho_A\, dx$, and $Q_V = \sum |P_i|^2 \ge 0$; Q_V vanishes at the mass points in W and $\int Q_V\, d\mu_\varpi = \dim V$. The Gram matrix $(\int e^{i(l' - l)\omega_0 x} d\mu_\varpi)_{ll'}$ is at least $\gamma\,\mathrm{Id}$, with Id the identity matrix, because $\varpi c_* \hat{\rho} \ge \gamma/(l_2 - l_1)$ on $[l_1, l_2]$, an interval of length $2\pi/\omega_0$; so $Q_V \le (n + 1)/\gamma$. Writing $Q_V = \sum_{|j| \le n} q_j e^{ij\omega_0 x}$ with coefficients q_j, the coefficient matrix, conjugated by the square root of the Gram matrix, is a projector of rank $\dim V$; hence $\sum |q_j|^2 \le (n + 1) \dim V/\gamma^2$.

With $\Psi = \sum_j q_j \Psi_{j\omega_0}$, every finite measure ν' on $[0, A]$ satisfies $\int Q_V \varpi\, d\nu' = \sum_y \Psi\, (\mathsf{P}\nu')$; with $\nu' = \rho_A - \pi$, $\dim V = \int Q_V \varpi\, d\pi + \sum_y \Psi(R_A - m)$, with absolute convergence because $|\Psi|$ grows at most

geometrically while $\mathsf{P}\nu'$ decays faster than any exponential. Now $\int Q_V \varpi\, d\pi \le (n+1)\epsilon_W/\gamma$, since $Q_V = 0$ at the mass points in W and $\varpi \le \epsilon_W$ off W; $\sum_{Y_W} |\Psi_{j\omega_0}||R_A - m| \le g_n(b_2) r_W^{1/2} \eta_W$; off Y_W the bound $|R_A - m| \le Mr$ holds, so that part is at most τ_n; and $\sum_j |q_j| v_j \le (\sum |q_j|^2)^{1/2} (2n+1)^{1/2} \max_j v_j$ for any $v_j \ge 0$. Dividing by $\dim V$ and using $(n+1)/\dim V \le r_b$ gives
$1 \le r_b \epsilon_W/\gamma + (r_b(2n+1))^{1/2}[g_n(b_2) r_W^{1/2} \eta_W + \tau_n]/\gamma$, so by (a) and (b),
$g_n(b_2)\eta_W \ge \gamma/(2(r_b(2n+1) r_W)^{1/2}) \ge \gamma/(2(r_b(2n_*+1) r_W)^{1/2})$. With $\log(1+z) \le z$ this gives
$(n\omega_0)^2 (b_2 - s_1)/(2(1-\vartheta)^2) \ge \log\bigl(\gamma/(2(r_b(2n_*+1) r_W)^{1/2} \eta_W)\bigr)$, that is,
$n \ge \frac{(1-\vartheta)(l_2-l_1)}{2\pi} \bigl(2\log(\cdots)/(b_2 - s_1)\bigr)^{1/2}$. As $n(k) < r_b k/(r_b - 1)$, it follows that $k > \frac{r_b - 1}{r_b} n \ge k_C$, a contradiction. For $k = 0$, $n = 0$ and $V = \mathbb{C}$ give $1 \le \epsilon_W/\gamma + [r_W^{1/2} \eta_W + \tau_0]/\gamma$, against (a), (b) and $k_C > 0$. □

A.7 Assembly of the proof

Fix $\xi \in (0,1)$ and $c_0 < c_0^{\max}(\xi)$, take $\delta \in (0, (1-\kappa_L)/2)$ and $r_b > 2$, choose $\vartheta \in (0, 1/2)$, and let $A \to \infty$. Put $L_c = A^{3/4}$, the margin around the mass points $M_a = u(A/\vartheta)^{1/2}$ with $u^2 = 8 \log A$, and the output margin $M_o = 2(2A\Lambda_0/\vartheta)^{1/2}$; both are $o(L_c)$, since $M_a = O((A \log A)^{1/2})$ and $M_o = O(A^{2/3})$. Place cores $[l_1, l_2]$ of length L_c in $[(\kappa_L + \delta)A, (1-\delta)A]$ with consecutive cores $2M_a + 1$ apart ($s_1 = \lceil \vartheta l_1 \rceil$, $s_2 = \lfloor \vartheta l_2 \rfloor$). Each core gets the window $W = [l_1 - M_a, l_2 + M_a]$; these windows are pairwise disjoint. Its output window is $Y_W = [b_1, b_2] = [l_1 - M_o, l_2 + M_o]$; output windows may overlap, since each lemma is applied to one window. Let h_W be the minimum of h over Y_W/A and put $\bar\eta_W^2 = 2\mathcal{L}(1, M) C_0 A^{1/2} e^{-\Lambda_0 h_W} + 2\sum_{y \in Y_W} T_A^2/r$. By Corollary 16, $\bar\eta_W \ge \eta_W$, and Lemma 17 holds with $\bar\eta_W$ in place of η_W, because its proof gives $g_n(b_2)\eta_W \ge \gamma/(2(r_b(2n_*+1) r_W)^{1/2})$. Uniformly over the windows:

1. The inequalities $T_A \le c_* A^{-1/2} e^{-(A-y)^2/(2A)}$ and $y \le (1-\delta)A + M_o$ bound the second term of $\bar\eta_W^2$ by $e^{-\delta^2 A/3}$, so $\log(1/\bar\eta_W) = (\Lambda_0/2) h_W - O(\log A)$, with $h_W \ge h_\delta := \min(h(\kappa_L + \delta/2), h(1-\delta/2)) > 0$ for large A. In particular $cA^{1/3} \le \log(1/\bar\eta_W) \le \Lambda_0$.

2. The estimates $c_* = \Theta(A^{-1/2})$ and $\hat\rho(x) \ge c_1/(x \log x)^{1/2}$ (Lemma 8) and $\varpi \ge 1/2 - o(1)$ on the core give $\gamma \ge c' L_c A^{-1/2} (A \log A)^{-1/2} = c' A^{-1/4} (\log A)^{-1/2}$. The Poisson Chernoff bounds at the two ends of W give $\epsilon_W \le \exp(-u^2 l/(2(l + M_a))(1 - o(1))) \le A^{-3}$ for large A. So (a) holds.

3. The count of Lemma 17 on a window is $f_W = \Theta(L_c (\log(1/\bar\eta_W)/l_2)^{1/2}) = \Theta(A^{5/12})$; take $n_* = n(\lceil f_W \rceil) = \Theta(A^{5/12})$, so $\Omega := n_* \omega_0 = \Theta(A^{-1/3})$. For $y > b_2$ one has $\Psi_0(y) \le P(\mathrm{Bin}(y, \vartheta) \le s_2) \le \exp(-(\vartheta y - s_2)^2/(2\vartheta y))$ and $g_{n_*}(y) \le \exp(\Omega^2 (y - s_1)/(2(1-\vartheta)^2))$. At $y = b_2$ the exponent satisfies $(\vartheta y - s_2)^2/(2\vartheta y) \ge \vartheta M_o^2/(2b_2) \ge 4\Lambda_0$, while $\Omega^2 b_2/(2(1-\vartheta)^2) \le (1 + o(1)) \log(1/\bar\eta_W)/(1-\vartheta) \le 2.1\Lambda_0$ (as $\vartheta < 1/2$). Beyond b_2 the derivative of the tail exponent is at most $-(\vartheta/2)(1 - (s_2/(\vartheta b_2))^2) = -\Theta(A^{-1/3})$, against the growth rate $\Omega^2/(2(1-\vartheta)^2) = \Theta(A^{-2/3})$. For $y < b_1$ one has $P(\mathrm{Bin}(y, \vartheta) \ge s_1) \le \exp(-(s_1 - \vartheta y)^2/(2(\vartheta y + (s_1 - \vartheta y)/3))) \le e^{-3\Lambda_0}$ at $y = b_1$ and smaller below, while $g_{n_*}(y) \le g_{n_*}(b_1) \le e^{2.1\Lambda_0}$. With $r \le c_*$ one gets $\tau_{n_*} \le e^{-\Lambda_0/2}$ for large A, so (b) holds with $r_b > 2$ fixed, and $\log\bigl(\gamma/(2(r_b(2n_*+1) r_W)^{1/2} \bar\eta_W)\bigr) = \log(1/\bar\eta_W) - O(\log A)$.

4. One has $b_2 - s_1 = l_2(1-\vartheta)(1 + O(A^{-1/4}))$ and $l_2 = x(1 + O(A^{-1/4}))$ for x in the core.

So Lemma 17 gives, on each window, $k_W \ge (1 - o(1))(1 - 1/r_b)(1-\vartheta)^{1/2} \frac{l_2 - l_1}{2\pi} (\Lambda_0 h_W/l_2)^{1/2}$, and $h_W = h(x/A)(1 - o(1))$ for x in the core, since h is continuous and positive on $[\kappa_L + \delta, 1 - \delta]$ and the window has relative length $o(1)$. The windows are disjoint, so $K(A) \ge \sum_W k_W$. The cores cover $[(\kappa_L + \delta)A, (1-\delta)A]$ except the spacings, of total length $O(A^{1/4} M_a) = O(A^{3/4} (\log A)^{1/2}) = o(A)$; dropping them from the integral below loses $O(A^{3/4} (\log A)^{1/2} A^{-1/3}) = o(A^{2/3})$. Summing as a Riemann sum,

$$K(A) \ge (1 - o(1))(1 - 1/r_b)(1-\vartheta)^{1/2} \frac{\Lambda_0^{1/2}}{2\pi} \int_{(\kappa_L + \delta)A}^{(1-\delta)A} \Bigl(\frac{h(x/A)}{x}\Bigr)^{1/2} dx$$
$$= (1 - o(1))(1 - 1/r_b)(1-\vartheta)^{1/2} \frac{c_0^{1/2}}{2\pi} A^{2/3} \int_{\kappa_L + \delta}^{1-\delta} \Bigl(\frac{h(f)}{f}\Bigr)^{1/2} df.$$

Divide by $A^{2/3}$, take the lim inf, then let δ tend to 0, r_b to ∞, ϑ to 0 and $c_0 \to c_0^{\max}(\xi)$, and maximize over ξ. The value of λ_M, the bounds of Proposition 11(a) and the constant are enclosed in ball arithmetic. At $\xi = 0.2411$ (an exact decimal), κ_L is isolated by bisection with signs proved by interval arithmetic, and $c_0^{\max} \in [0.06490794803, 0.06490794804]$. The integral $\mathcal{I}$ is bounded below by a rigorous quadrature over $[\kappa_L + 10^{-8}, 1 - 10^{-8}]$, where the integrand is analytic, and above by adding the two end pieces with the bound $\kappa_L^{-1/2}$ on the integrand; this gives $\mathcal{I} \in [1.3824928, 1.3824930]$ and $c_0^{\max}(\xi)^{1/2}\mathcal{I}(\xi)/(2\pi) \in [0.0560572, 0.0560573]$. Since c_P is a maximum over ξ, it follows that $c_P > 0.0560$. □

A.8 A spontaneous count

Fix $\lambda > 0$, let π be the capacity-achieving input of $Y \sim \text{Poisson}(X)$ with $\lambda \le X \le U := \lambda + A$, and let C be its capacity. Fact (F1) holds for this channel, with $[\lambda, U]$ in place of $[0, A]$ [3] [12, Lemma 1]; (F3) and (F4) hold for every input law [12, Lemma 3]; (F2) is not used below. Define r, c_* and g as before, with this C. Lemma 4 and the identity of Proposition 5 do not depend on the input range, so the optimality conditions read $E_x g(Y) \ge \delta_0(x)$ on $[\lambda, U]$, with equality at every mass point. The value $g(0) = 1$ is not available. It was used in Proposition 10 when g is maximal at 0, in Proposition 11 at $y_1 = 0$, and in Lemma 15(i) through $m(0) = e\,c_*$. The lemma below replaces the first use.

Lemma 18 (the value at zero)**.** *For every $\lambda \ge 0$, with $m_0 = E[X \mid Y = 0]$ and $m_1 = E[X \mid Y = 1]$, one has $g(0) \le 1 - m_0 + m_0 \log(m_1/m_0)$. If g is maximal at 0, then $g(0) \le 1 + 2e^{-1-\gamma_E} < 1.42 < \lambda_M$.*

Proof. By (F3), $G(1) - G(0) = \log m_0$, so convexity of G gives $G(y) \ge G(0) + y \log m_0$ on $\mathbb{N}_0$; with $g = G - \varphi + C + 1$ and $\varphi(0) = 0$ this gives $g(y) \ge g(0) + y \log m_0 - \varphi(y)$. Take E_{x_j} at a mass point x_j; as in Lemma 9, $\delta_0(x_j)$ cancels and $g(0) \le 1 - m_0 + B(x_j, m_0)$. Average over the posterior of X given $Y = 0$, which has mean m_0. The posterior given $Y = 1$ is that posterior tilted by x, so $E[B(X, m_0) \mid Y = 0] = m_0 E[\log(X/m_0) \mid Y = 1] \le m_0 \log(m_1/m_0)$ by Jensen's inequality. If g is maximal at 0, then $g(2) \le g(0)$; since $G(2) - G(0) = \log m_0 + \log m_1$ and $\varphi(2) = \psi(1) + \psi(2) = 1 - 2\gamma_E$, this gives $\log m_1 \le 1 - 2\gamma_E - \log m_0$. Hence $g(0) \le 1 - 2\gamma_E m_0 - 2m_0 \log m_0$, and the right side is largest at $m_0 = e^{-1-\gamma_E}$, where it equals $1 + 2e^{-1-\gamma_E} = 1.4131\ldots$. □

With this lemma the proof of Proposition 10 holds for $\lambda > 0$, since its case $y_0 \ge 1$ uses only the mass points, the posteriors and the values of g at $y_0 \pm 1$; so $m \le Mr$ on $\mathbb{N}_0$. Proposition 11 holds for every integer $y_1 \in [\lambda, U]$: its proof uses the optimality inequality at the feasible point $x = y_1$, the case $y_1 = 0$ does not arise, and the bounds of its part (a) do not depend on the channel. Corollary 12 follows with $[\lambda, U]$ in place of $[0, A]$ in (LM1). For the integers $y < \lambda$, put $y_1 = \lceil \lambda \rceil$; convexity of G with the slope $\log E[X \mid Y = y_1 - 1] \le \log U$ gives $g(y) \ge -\lambda_\ell - y_1(\gamma_E + \log U)$, so $|g(y)| = O(\log A)$ there.

Take ρ to be the restriction of ρ_∞ to $[\lambda, U]$; Lemma 7(iv) holds for it, since its proof uses only that $\rho/\rho([\lambda, U])$ is a feasible input. Then $\mathsf{P}\rho = r - T_U - L_\lambda$ on $\mathbb{N}_0$, where T_U is T_A with U in place of A and $L_\lambda = \mathsf{P}(\rho_\infty \mathbf{1}_{[0,\lambda)})$. As $p(y \mid x)e^x \le \lambda^y/y!$ for $x \le \lambda$ and $\int e^{-x}\rho_\infty(dx) = r(0) = c_*$, one has $L_\lambda(y) \le c_* \lambda^y/y!$. The proof above now runs with U in place of A in the weight χ, the window and the cores, with three changes:

1. In Step 1 of Proposition 13, $F = r - T_U - L_\lambda - m$, and the identity gains the term $-\tilde{L}$ with $\tilde{L} = \sum_{y \le U} \chi g L_\lambda$. Below x_L one has $\chi \le 1$ and $|g| = O(\log A)$; on and above x_L the weight satisfies $\chi \le e^{\Lambda_0}$, while $\lambda^y/y! \le (e\lambda/y)^y$ decays faster than any exponential. So $|\tilde{L}| = O(c_* \log A)$, which enters ϵ_{out}. The part of $\tilde{T}$ below λ is bounded in the same way, since $\chi(y) \le (\lambda/U)^{2\sigma} e^{tU}$ there; the rest of $\tilde{T}$ is bounded as in Step 3.

2. In Step 6, $\|\mathcal{P}\|$ gains the term $(\sum_y \chi L_\lambda^2)^{1/2} = O(c_*)$, which enters b; in Lemma 17 and the assembly, $T_U + L_\lambda$ replaces T_A in $\bar{\eta}_W$, and L_λ is negligible on the output windows.

3. In Lemma 15(i), $m(0) = e^{g(0)} c_* \le M c_*$, so $|d_0| \le (1 + M)c_* + 2e^{\Lambda_0 - x_L}$. In (iv), $|F| \le Mr$ on $y \le U$ because $0 \le T_U + L_\lambda \le r$; in (iii), $y_A = \lfloor U \rfloor$ lies in $[\lambda, U]$ for $A \ge 1$. Lemma 8(ii) sums over the integers of $[\lambda, U]$ and still gives $c_* = \Theta(A^{-1/2})$.

None of these changes alters δ', so its limit and the conditions $b = O(A^{1/4})$ and $\epsilon_{\text{out}} = O(A^{1/2})$ stand. The assembly gives $K(A, \lambda) \geq (1 - o(1))(c_P - \varepsilon)\, U^{2/3}$ for every $\varepsilon > 0$, and $U^{2/3}/A^{2/3} \to 1$, which proves the theorem for $\lambda > 0$. □

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